Variant Configuration (VC)

The case study describes the creation of a configurable product and its use in an integrated discrete manufacturing process for a make-to-order customer. It shows the possibilities of simplifying master data by adding characteristics, which allows a product to have different characteristics.

Product

S/4HANA 2023
Global Bike

Fiori 3.0

Level

Undergraduate
Graduate
Beginner

Focus

Variant Configuration

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Version

4.3

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MOTIVATION

After you have only displayed data such as bills of materials and routings in the
-control exercises, this case study is about creating a configurable product and using it in an integrated process from sales order to production.

You will create new characteristics, classes, and material master records and relate them to each other to ensure configurability from product selection to BOM explosion.

PREREQUISITES

Before you use this case study, you should be familiar with navigation in the SAP system.

In order to successfully work through this case study, it is not necessary to have finished the PP exercises (PP 1 through PP 6). However, it is recommended.

NOTES

This case study uses the Global Bike data set.

SAP University Alliances

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Process Overview

Learning Objective Create and understand variant configuration and make-to-order production. **Time** 120 min

Scenario To handle a complete customer make-to-order process, you will take on different roles within Global Bike, such as warehouse worker and production manager. Overall you will work in the departments Production Planning (PP), Materials Management (MM) and Sales and Distribution (SD).

Employees involved

Jun Lee (Production Manager)
Lars Iseler (Shop Floor Worker 2)
Susanne Castro (Goods Receipt Clerk)
Sanjay Datar (Warehous Employee)
Jamie Shamblin (Controller)

You start preparing the variant configuration by defining characteristics and classes. These are used to create a configurable product that is supplemented by a maximum bill of material and a routing. Once variant configuration is complete, the configurable material is ordered as part of a sales order and produced as make-to-order.

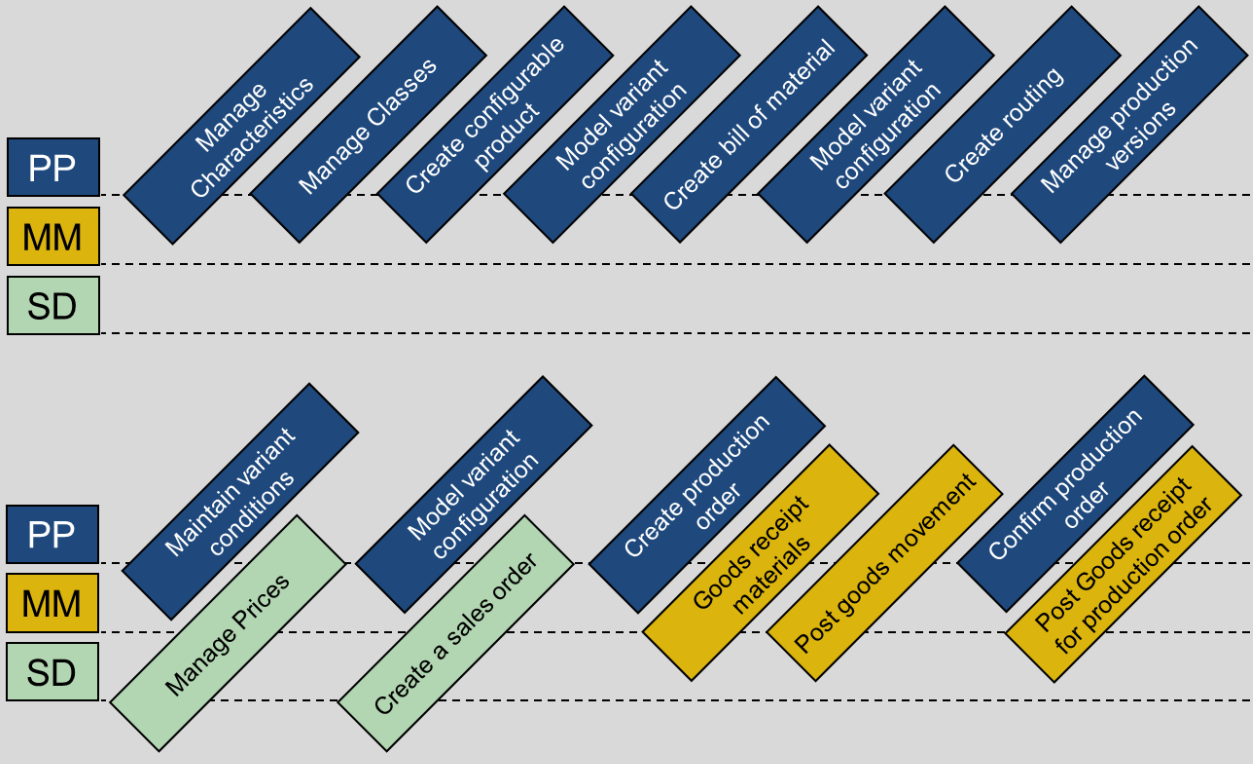


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Step 1: Manage Characteristics

Task Prepare a characteristic for variant configuration. **Time** 10 min

Short Description To be able to configure a Global Bike's Deluxe Touring Bike, features must be prepared. These can then be assigned to a product.

Name (Position) Jun Lee (Production Manager)

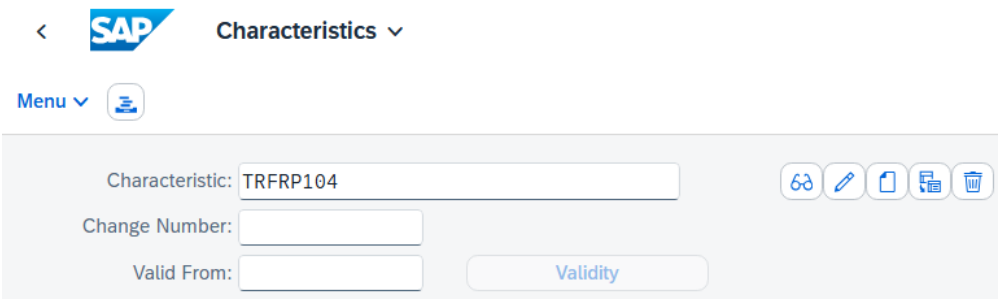
To create a new characteristic, use the *Manage Characteristics* app in the *Production Planning and Execution* area on the *Variant Configuration* page in the *Production Manager* role.

Start



Enter **TRFRP###** (replace ### with your three-digit number) as the new Characteristic.

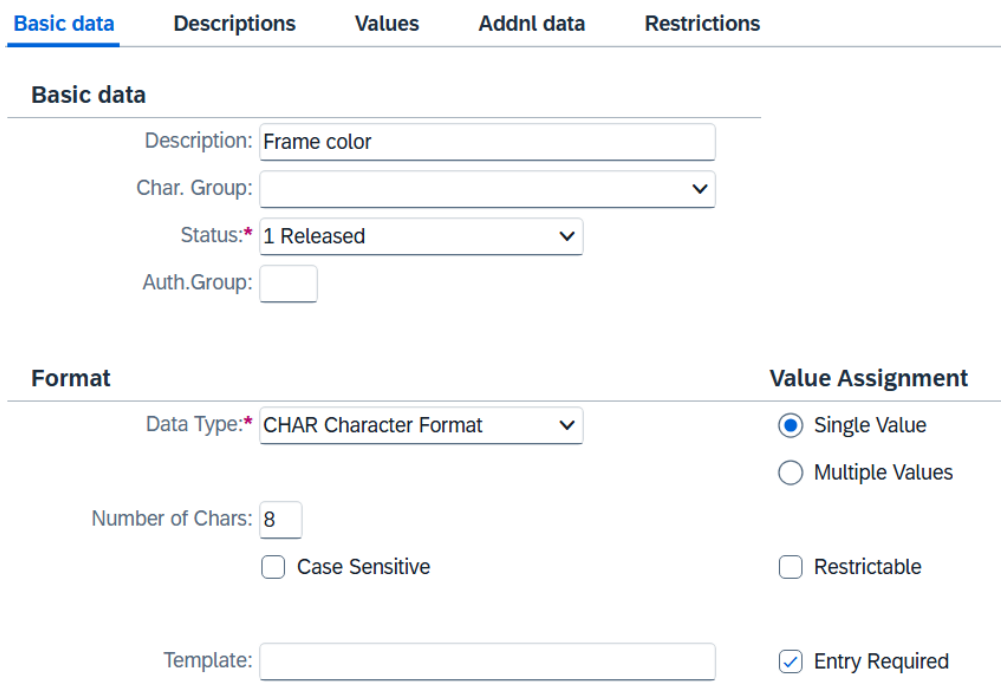
TRFRP###



Press to create the Characteristic.

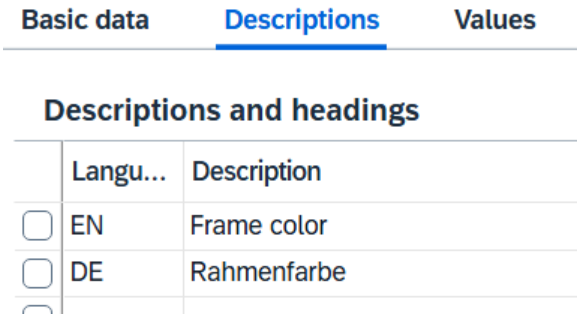
In the *Basic data* tab, enter **Frame color** as a *Description* Check that **Released** is selected as the *Status*. Then select **CHAR** as the *data type*, enter **8** as the *number of Chars* and select the options **Entry required**.

Frame color
Released
Char
8
Entry required



Switch to the *Descriptions* tab. Add the *description* **Rahmenfarbe** for the **DE** language.

DE
Rahmenfarbe



Switch to the *Values* tab. For the Char.Value, we will use the existing Touring Bike frames as a guide:

Touring Bike Frame - Black - TRFR1###

Touring Bike Frame - Silver - TRFR2###

Touring Bike Frame - Red - TRFR3###

The different colors can be recognized by the consecutive number. This principle is supernumerary. To do this, enter the following characteristic values:

Black
Silver
Red

Char. Value	Description
TRFR1###	Black
TRFR2###	Silver
TRFR3###	Red

Note The characteristic values and the material numbers of the frames are identical in this example. However, they are two values that are detached from each other and can also differ from each other.

Navigate to *menu* → *Extras* → *Change language*. A popup will open, select **DE** and press ✓ to accept the selection.

DE

Now enter the German Description:

Char. Value	Description
TRFR1###	Schwarz
TRFR2###	Silber
TRFR3###	Rot

Schwarz
Silber
Rot

Press Save to save your entries.

Click SAP to return to the SAP Fiori Launchpad.

☐

Step 2: Manage Classes

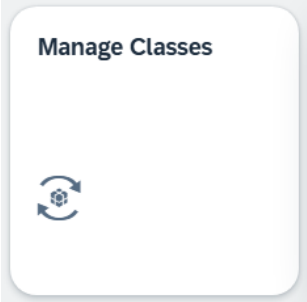
Task Create a class and assign a characteristic. **Time** 5 min

Short Description Classes are used to group one or more characteristics together and can then be assigned to a material. It is also possible to use classes in BOMs to then select linked materials depending on the selection.

Name (Position) Jun Lee (Production Manager)

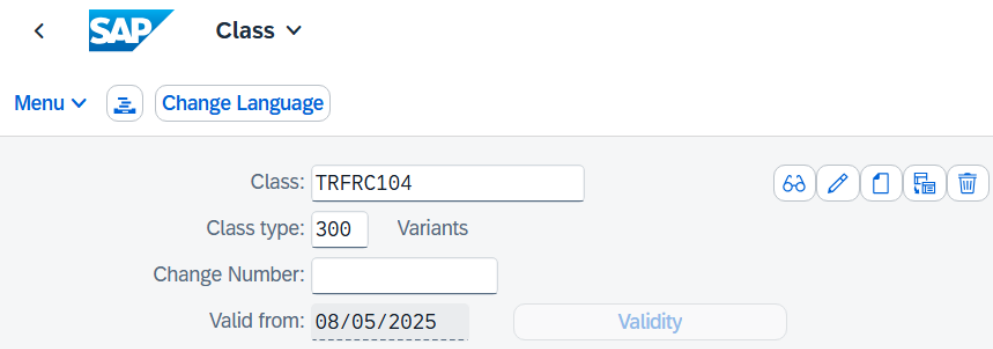
To create a class, use the *Manage Classes* app in the *Production Planning and Execution* area on the *Variant Configuration* page in the *Production Manager* role.

Start



Enter **TRFRC###** as the *class* and **300** as the *class type*.

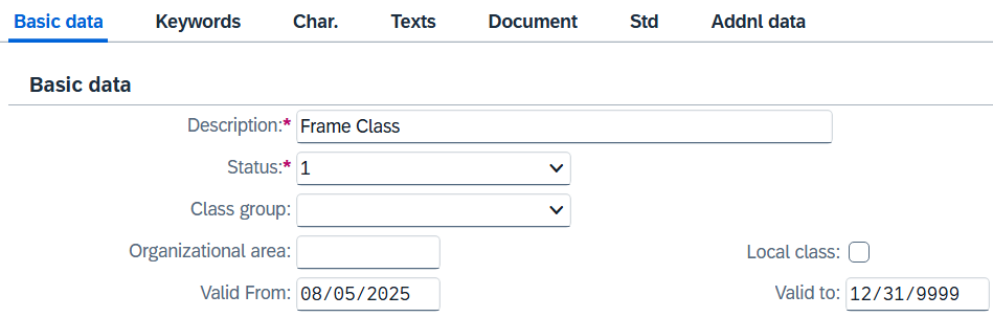
TRFRC###
300



Press to create the Characteristic.

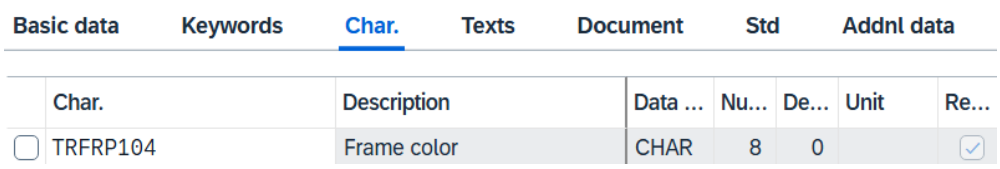
In the *Basic data* tab, enter **Frame class** as the *description*. Check that **1** is selected as the *status*.

Frame class
1



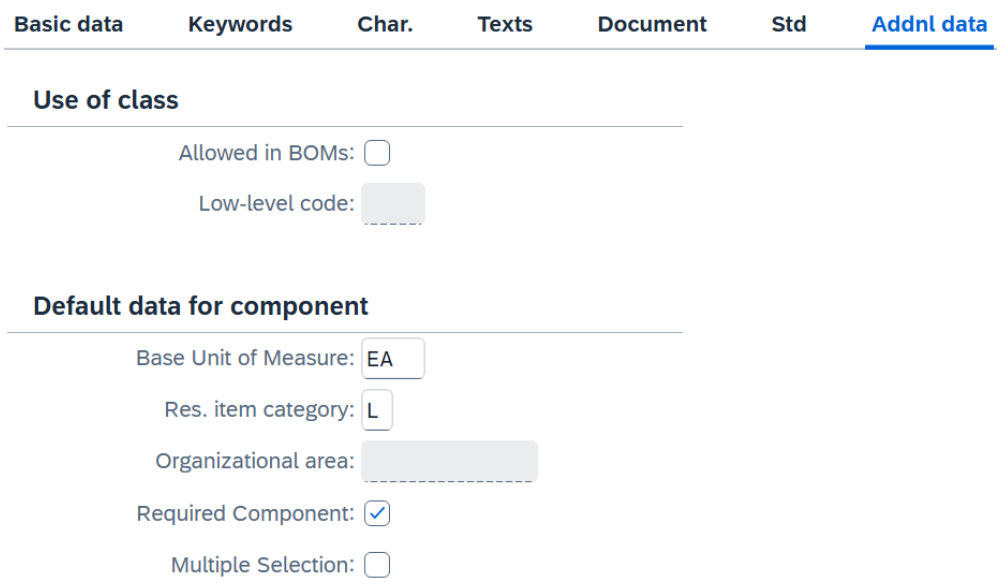
Switch to the *Char.* tab and enter the newly created characteristic **TRFRP###**. Press Enter.

TRFRP###



Switch to the *Additional data* tab. Enter **EA** for *Base unit of measure* and **L** for *Res. item category*. Then select **Required component**.

EA
L
Required component



Press Save to save your entries.

Click SAP to return to the SAP Fiori Launchpad.

Step 3: Create Configurable Product

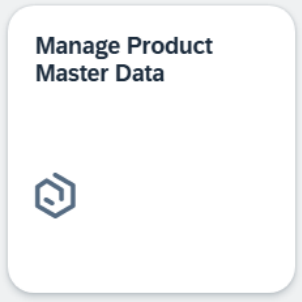
Task Create a configurable product. **Time** 10 min

Short Description Based on a Deluxe Touring Bike, a configurable material will be created that includes the previously defined class. This product will then map all possible Deluxe Touring Bike versions. However, for the sake of simplicity, the new product will only be created in the HD00 plant.

Name (Position) Jun Lee (Production Manager)

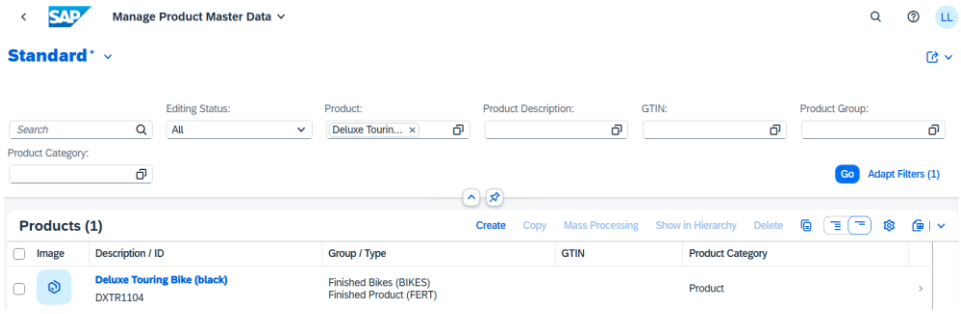
To create a configurable product, use the *Manage Product Master Data* app in the *Production Planning and Execution* area on the *Variant Configuration* page in the *Production Manager* role.

Start



Enter **DXTR1##** in the *Product* field and press Go. The corresponding data record is displayed.

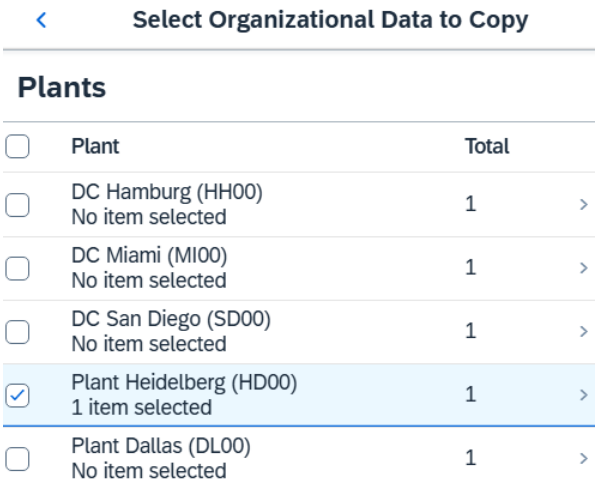
DXTR1###



Select the entry and press Copy, the *Select organizational data to copy* popup opens.

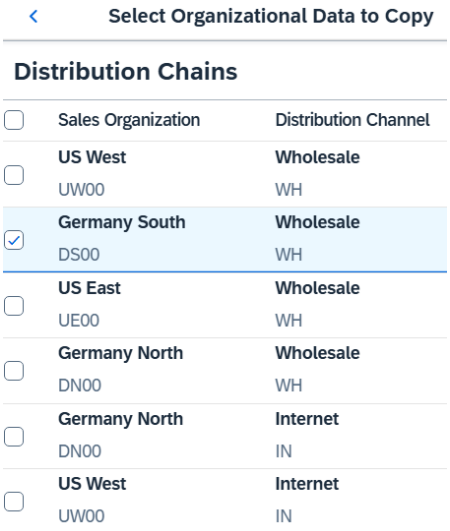
Click on the entry *Plants*, all plants defined for DXTR1## are listed. Deselect all entries and select only the entry **Plant Heidelberg (HD00)**.

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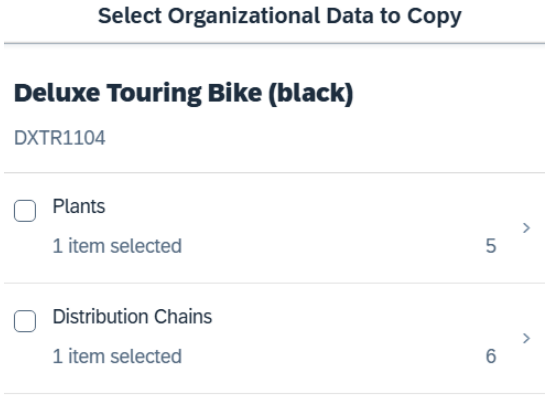


Press < to navigate back and then press *Distribution Chains*.

Deselect all entries again and select only the entry **Germany South (DS00) - Wholesale (WH)**.



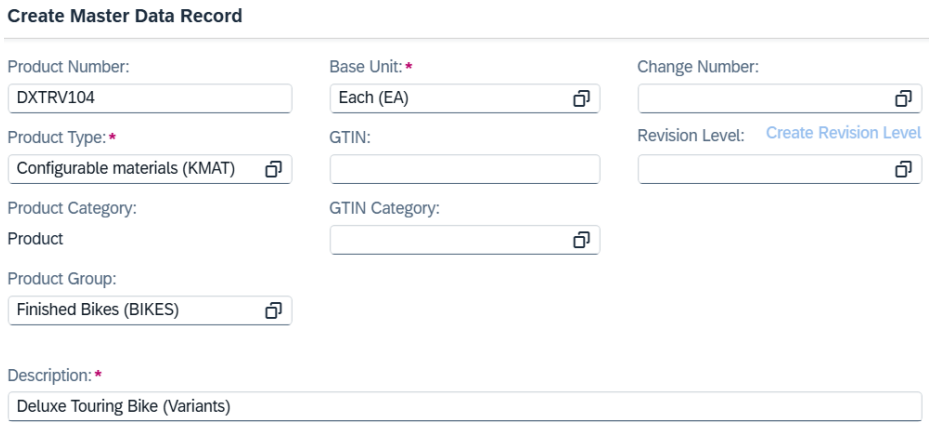
Press < to navigate back. In the overview, exactly one position is now selected for each entry.



Press OK to accept the selection.

Enter **DXTRV###** for *product number*, select **Configurable Materials (KMAT)** as *product type* and change *description* to **Deluxe Touring Bike (Variants)**.

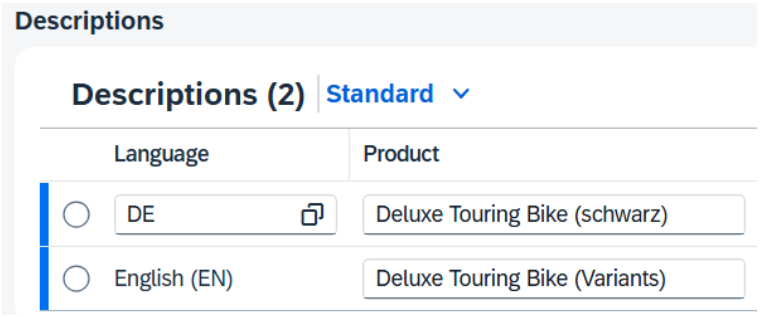
DXTRV###
KMAT
Deluxe Touring Bike (Variants)



Confirm your entries with OK. The copied master data record is now displayed.

In the *Descriptions* section, delete the entry for the languages *RU* and *FR* by selecting them and clicking on Delete. Then change the entry for the *DE* language to **Deluxe Touring Bike (Varianten)**.

Deluxe Touring Bike (Varianten)



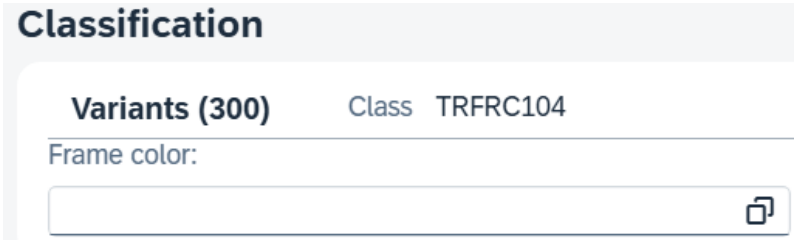
Select the *Classification* area, the window will automatically scroll to the correct position.

Click on Click here to assign a class., a selection window opens. Search for your class **TRFRC###**.

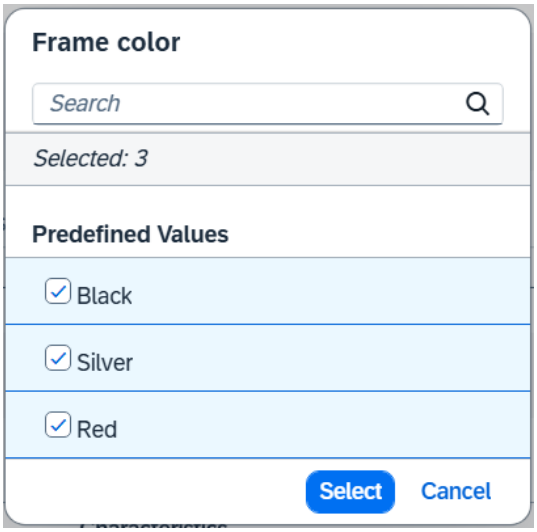
TRFRC###



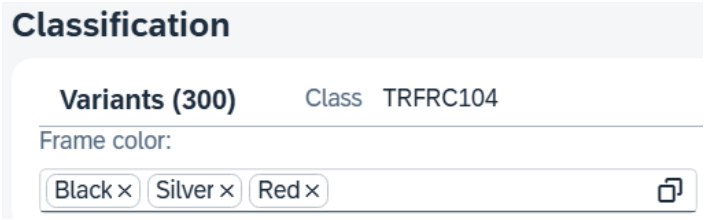
Select your class, it will be added to the master record.



So that a selection can be made later, the desired values of the characteristic must still be selected. To do this, press to open the input help. Then select all the predefined values.

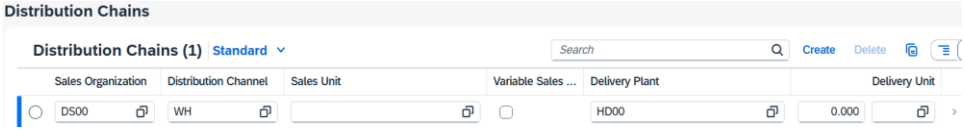


Press Select to accept the values.



Select the *Distribution Chains* section, the window will automatically scroll to the correct position.

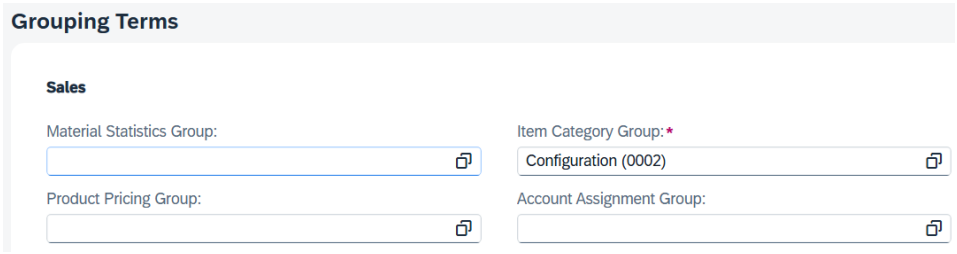
Only the pre-selected distribution chains *DS00 - WH* was taken over.



Click on › at the end line to open the details.

Select the *Grouping terms* area and choose **Configuration (0002)** as the *item category group*.

Configuration (0002)

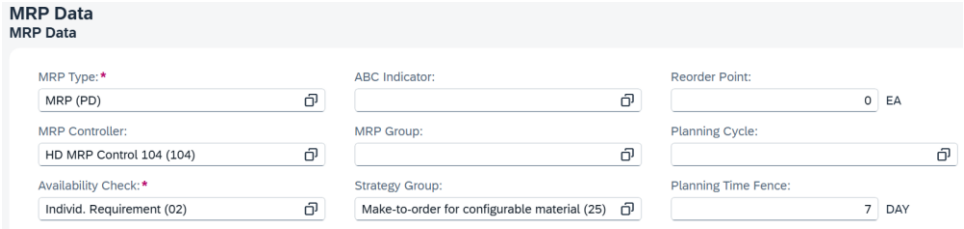


Press Apply, you will return to the general record.

Open the details of the HD00 plant. The *Plants* area is located directly below the *Distribution Chains* area. To do this, click › at the end line.

Select the *MRP Data* area. Change the *MRP type* to **Planned Disposition (PD)** and select **Make-to-order for configurable material (25)** as the *strategy group*.

Planned Disposition
Make-to-order for configurable material



Scroll down a bit to the *Procurement* section. Select **In-house production** (E) as the *procurement type*. This is mandatory and can no longer be changed, since configurable material is always created in in-house production.

In-house production



After clicking on Apply, you will return to the general record. To create the new material please press the Create button.

Click SAP to return to the SAP Fiori Launchpad.

Step 4: Model Variant Configuration

Task Create a configuration profile. **Time** 5 min

Short Description The configurable product requires a configuration profile to resolve the characteristics of the class at runtime and offer them as choices.

Name (Position) Jun Lee (Production Manager)

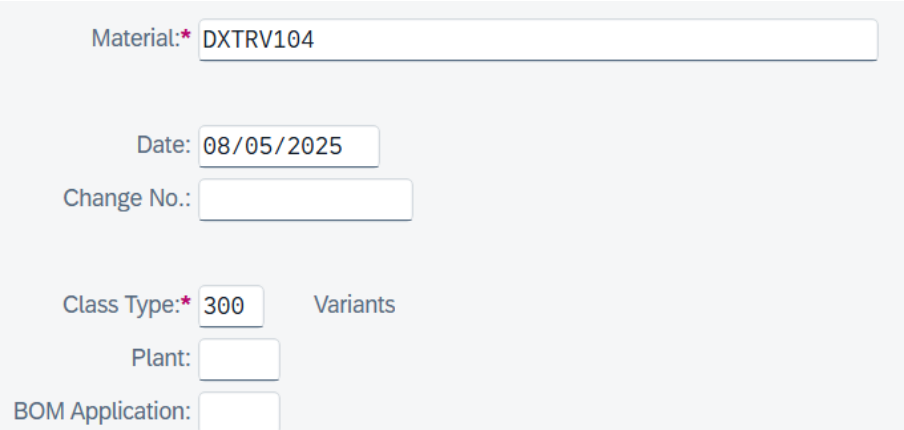
To model variant configuration, use the *VC Modeling Environment* app in the *Production Planning and Execution* area on the *Variant Configuration* page in the *Production Manager* role.

Start



Enter your configurable material **DXTRV###** for *Material* and verify that **300** is selected as the *class type*.

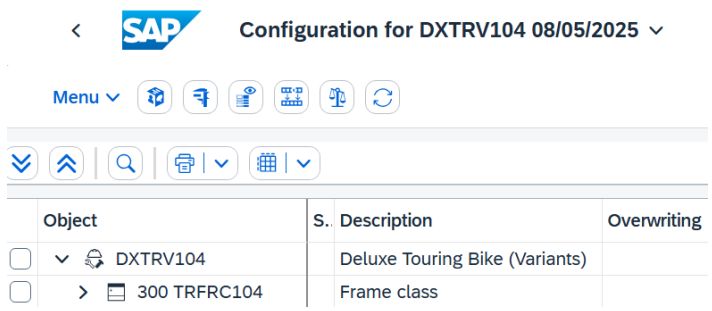
DXTRV###
300



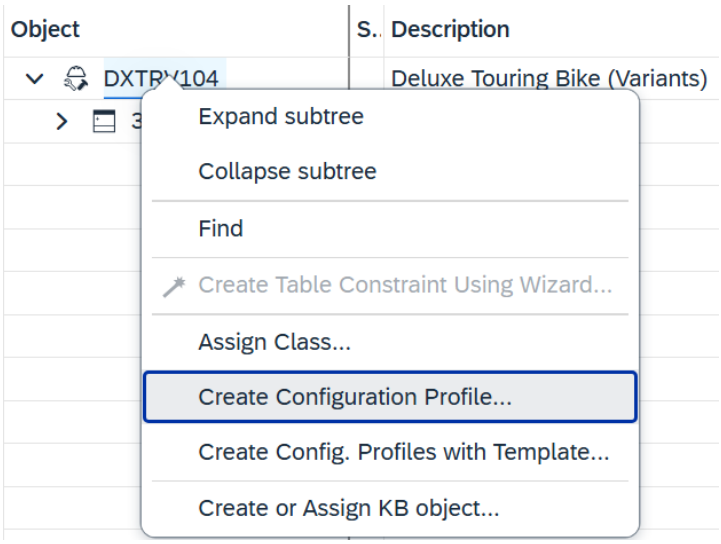
Press to enter the model.

You will be shown the configuration for your configurable material.

The *DXTRV##* object represents the Configurable product. The class *(300 TRFRC##)* is located below it. You can also expand the individual elements to view their structure.

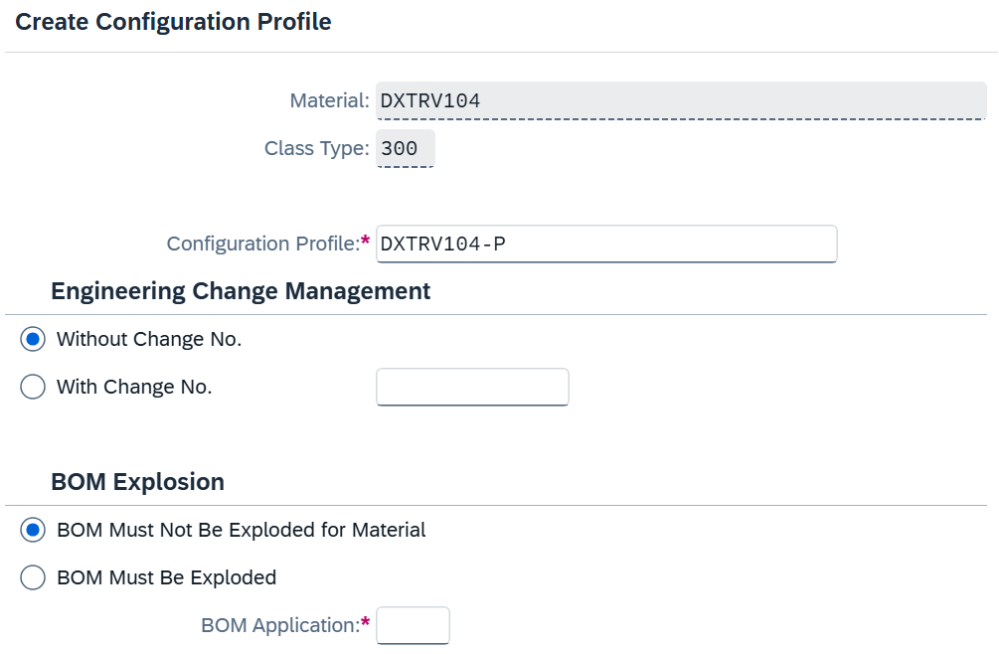


Right-click on the *DXTRV###* entry to open the corresponding context menu.



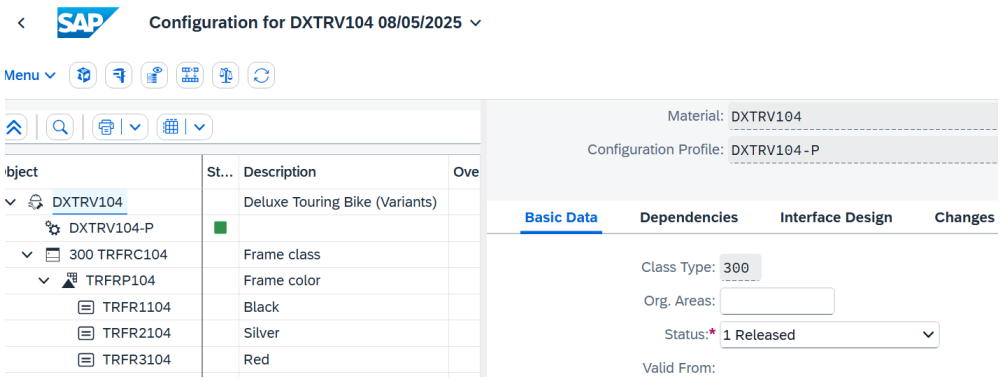
Select *Create configuration profile*, a new window opens. Enter **DXTRV###-P** for *Configuration profile*.

DXTRV###-P



Press ✓ to apply the profile.

The new configuration profile opens automatically and is also visible in the navigation on the left side of the screen with a green status.



Further settings in the configuration profile are not necessary.

Press Save to save your changes.

Click SAP to return to the SAP Fiori Launchpad.

Step 5: Create Bill of Material

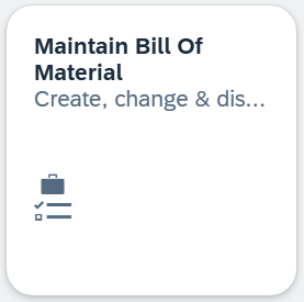
Task Create maximum bill of material. **Time** 5 min

Short Description For the configurable Deluxe Touring Bike, a maximum bill of materials must be defined that includes all possible components.

Name (Position) Jun Lee (Production Manager)

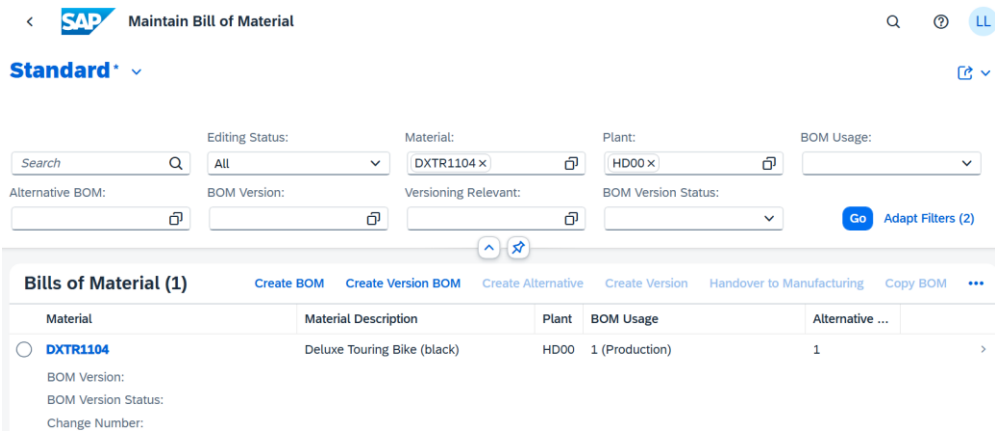
To customize the BOM, use the *Maintain BOM* app in the *Production Planning and Execution* area on the *Variant Configuration* page in the *Production Manager* role.

Start



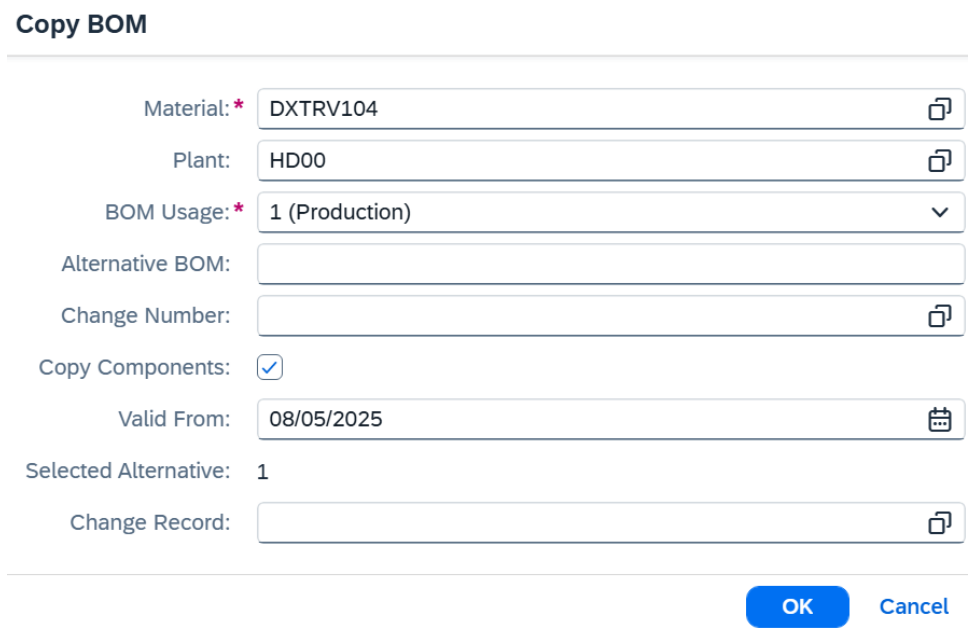
Enter **DXTR1###** in the *Material* field and **HD00** in the *Plant* field. After pressing Go, the corresponding bill of material will be displayed.

DXTR1###
HD00



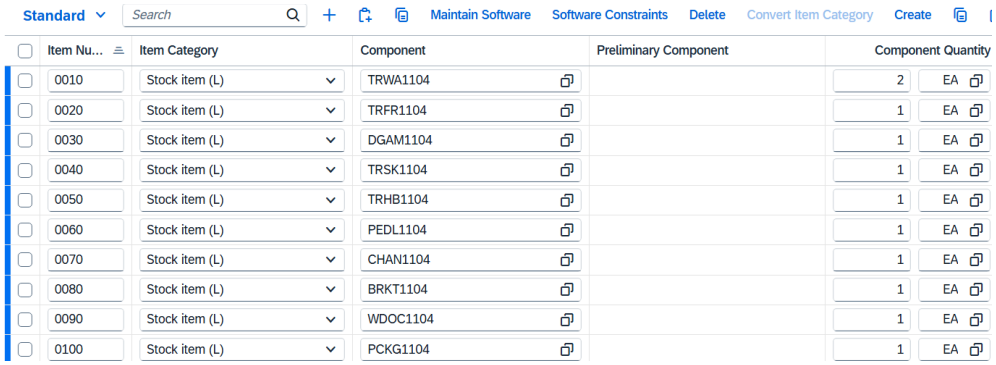
Select the entry and press Copy BOM. In the window that opens, enter **DXTRV###** for *material* and **HD00** for *plant* and select **1 (Production)** as the *BOM usage*.

DXTRV###
HD00
1



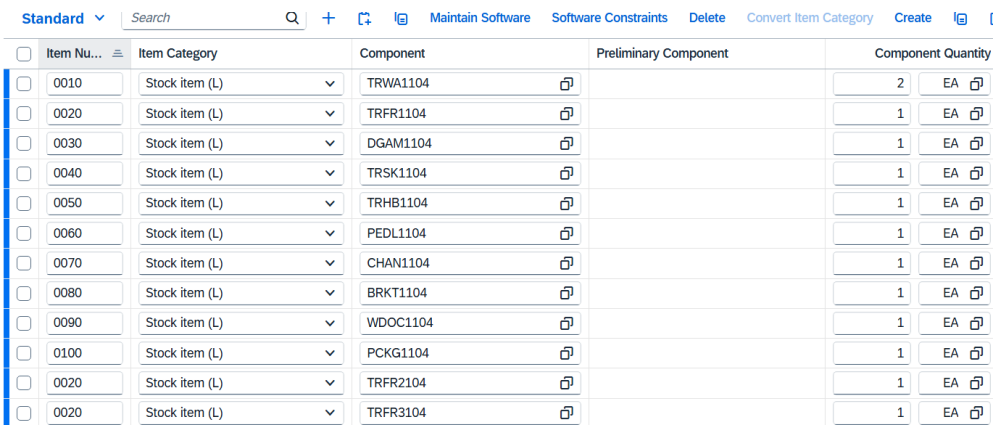
Press OK to accept the entries.

You will see the bill of materials with its components.

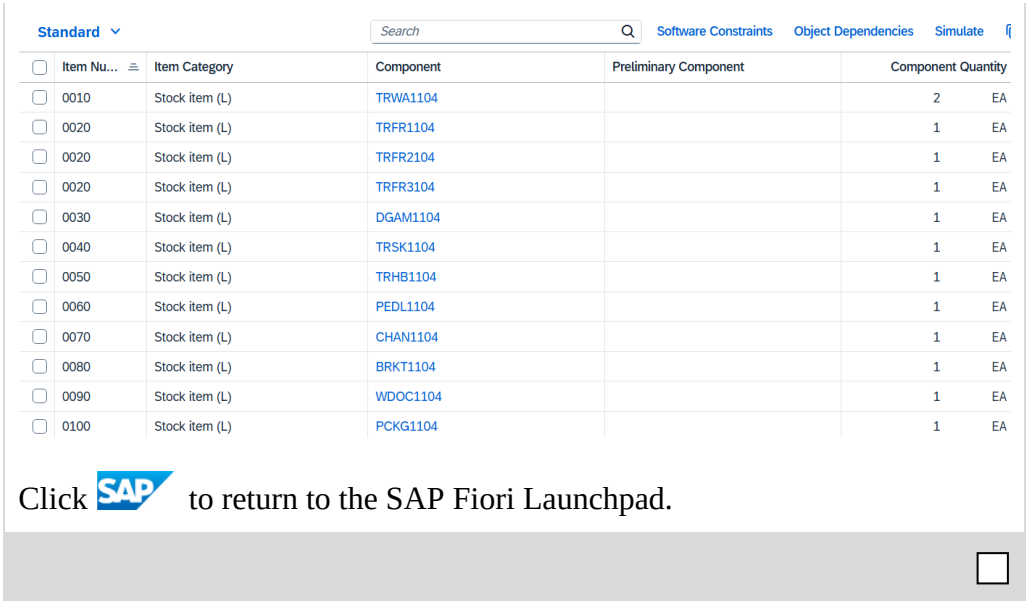


To match this to the configurable product, change the free positions *0110* and *0120* to **0020**, then **TRFR2###** and **TRFR3###** as *component* with a component quantity of **1 EA** each.

0020
TRFR2###
1 EA
0020
TRFR3###
1 EA



Press Create to save the new parts list. After saving, the new entries are also displayed sorted by item number.



Item Nu...	Item Category	Component	Preliminary Component	Component Quantity
0010	Stock item (L)	TRWA1104		2 EA
0020	Stock item (L)	TRFR1104		1 EA
0020	Stock item (L)	TRFR2104		1 EA
0020	Stock item (L)	TRFR3104		1 EA
0030	Stock item (L)	DGAM1104		1 EA
0040	Stock item (L)	TRSK1104		1 EA
0050	Stock item (L)	TRHB1104		1 EA
0060	Stock item (L)	PEDL1104		1 EA
0070	Stock item (L)	CHAN1104		1 EA
0080	Stock item (L)	BRKT1104		1 EA
0090	Stock item (L)	WDOC1104		1 EA
0100	Stock item (L)	PCKG1104		1 EA

Click SAP to return to the SAP Fiori Launchpad.

Step 6: Model Variant Configuration

Task Create object dependencies for the super bill of material.. **Time** 15 min

Short Description In order for the characteristic selection to have a dynamic effect on the maximum bill of material, the characteristic values must be linked to the components of the bill of material. Depending on the selected characteristic, the appropriate touring bike frame is then used.

Name (Position) Jun Lee (Production Manager)

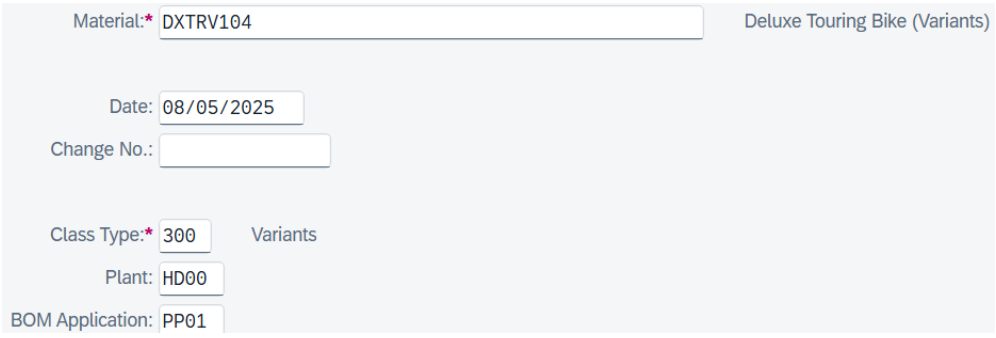
To model variant configuration, use the *VC Modeling Environment* app in the *Production Planning and Execution* area on the *Variant Configuration* page in the *Production Manager* role.

Start



Enter your configurable material **DXTRV###** for *Material*. Check that **300** is selected as the *class type* and enter **HD00** for *Plant*, and **PP01** for *BOM Application*.

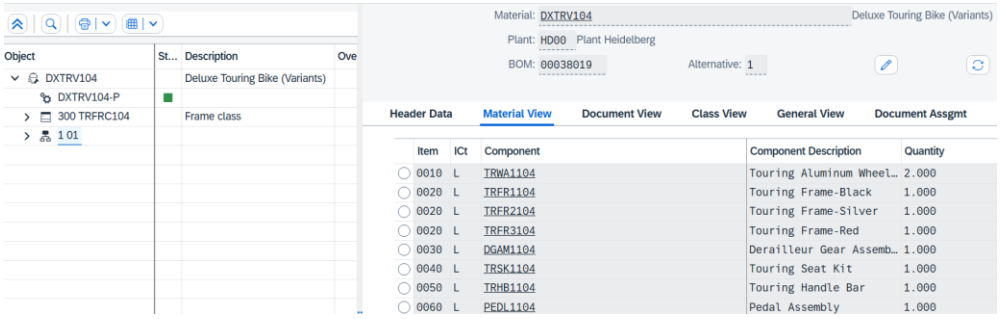
DXTRV###
300
HD00
PP01



Press to enter the model.

The configuration for your configurable material is displayed. In addition to the frame class and the configuration profile created, the list *(1 01)* is now also displayed.

Double-click on the list *(1 01)* in the object list to open its details.



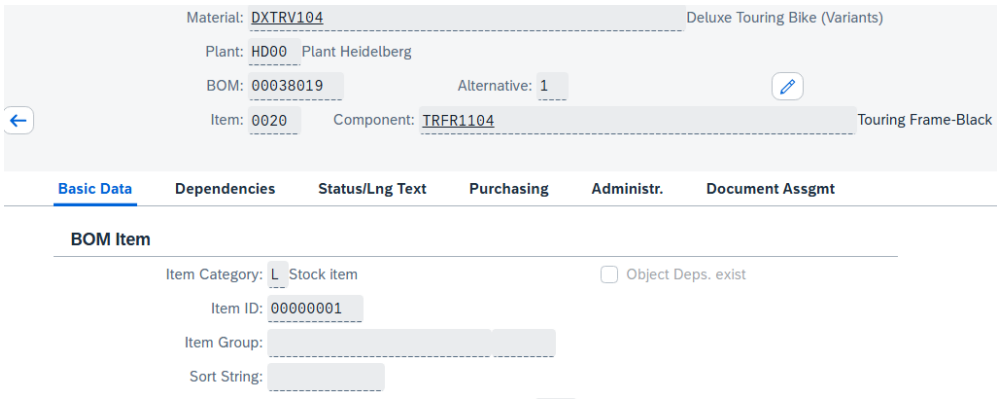
Press  to activate the edit mode.

A message appears. Leave the selection at *Without change no.* and confirm with ✓.

Now expand the list *(1 01)* in the object list to see the individual items.

Object	St...	Description
DXTRV104		Deluxe Touring Bike (Variants)
DXTRV104-P	■	
300 TRFRC104		Frame class
1 01		
0010 L TRWA1104		Touring Aluminum Wheel Assembly
0020 L TRFR1104		Touring Frame-Black
0020 L TRFR2104		Touring Frame-Silver
0020 L TRFR3104		Touring Frame-Red
0030 L DGAM1104		Derailleur Gear Assembly
0040 L TRSK1104		Touring Seat Kit
0050 L TRHB1104		Touring Handle Bar
0060 L PEDL1104		Pedal Assembly
0070 L CHAN1104		Chain
0080 L BRKT1104		Brake Kit
0090 L WDOC1104		Warranty Document
0100 L PCKG1104		Packaging

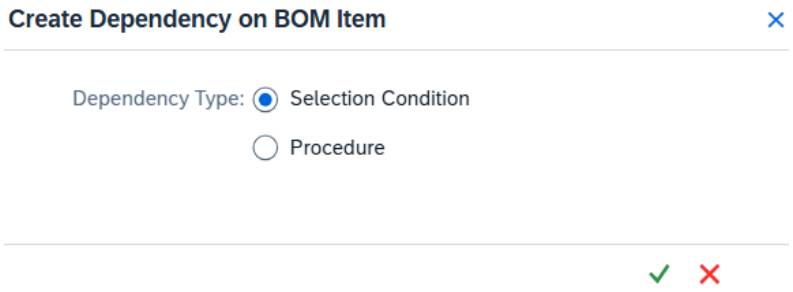
Double-click on 0020 L TRFR1### in the object list to open the details of the position.



Switch to the *Dependencies* tab. There are currently no entries there. Press Local to create a local object dependency.

Note Object dependencies are used to link information and objects with each other, to make them dependent on each other, or to exclude each other. In this way, for example, conditions can be ensured during configuration or derivations can be made. Local dependencies apply only to one object, whereas global dependencies can be reused.

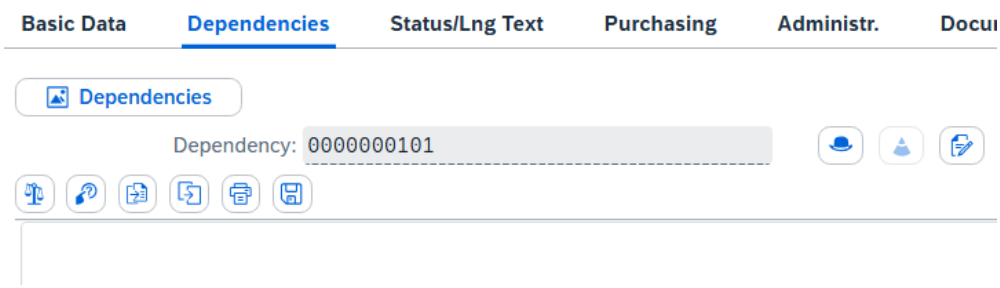
The *Create Dependency on BOM Item* window opens.



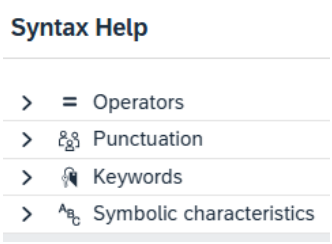
Confirm with ✓ the *dependency type* **Selection conditions**.

Selection conditions

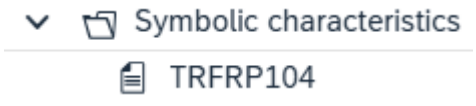
The still empty dependencies editor opens. The first relationship is created piece by piece to develop an understanding of it. You can then enter the other relationships directly.



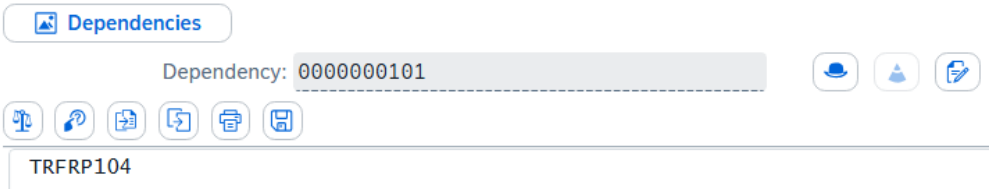
The dependencies editor provides context-sensitive input help to create dependencies. Press to access the syntax help.



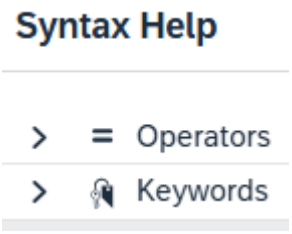
You will see the different options Operators, Punctuation, Keywords and Symbolic characteristics. Expand the entry for the Symbolic characteristics.



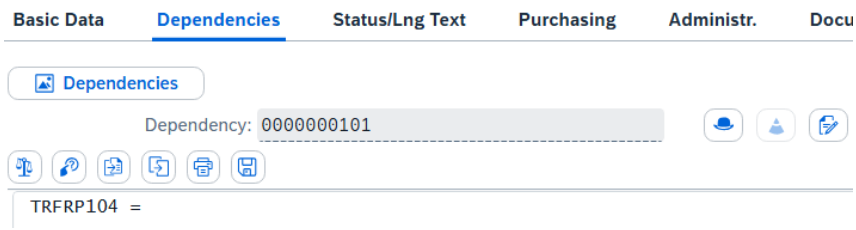
You will see your characteristic *TRFRP##*. Select it by double-clicking, it will be transferred to the dependency editor.



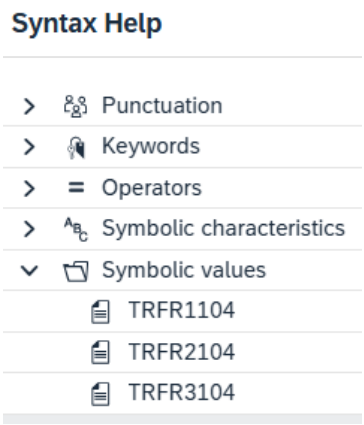
Press again to call the syntax help. This time you will be offered a small selection of possibilities, since at the point only an operator or a keyword can follow for a syntactically correct relationship.



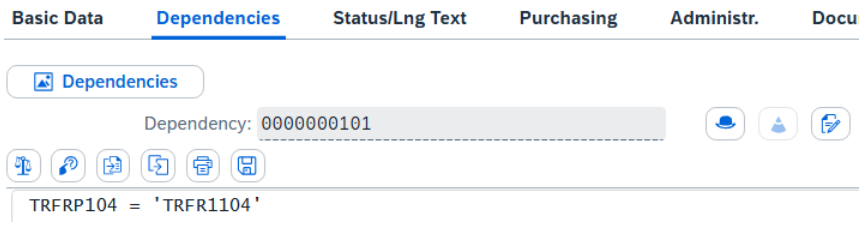
Expand operators and insert = (equal sign) by double-click.



Press again to call the syntax help. In addition to the options listed at the beginning, symbolic values are now available. Expand them.



You will see characteristic values *TRFR1###*, *TRFR2###* and *TRFR3###*. Select *TRFR1###* by double-clicking it to take it over to the dependency editor.

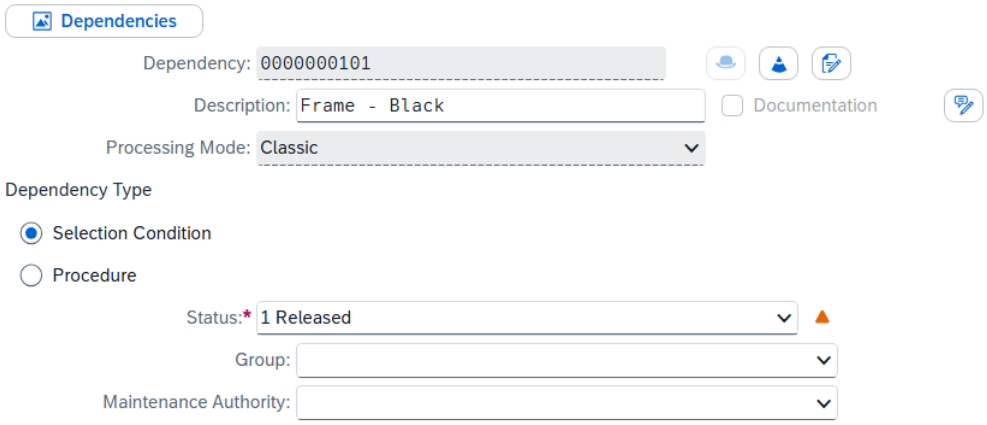


The rule is now complete. If the characteristic TRFR1## is used, the component TRFR1## is selected for the BOM.

Press to switch to the basic data of the dependencies.

Enter **Frame - Black** as the *description* and select **1 Released** as the *status*.

Frame - Black



Note With you can switch back to the editor.

Repeat the procedure for 0020 L TRFR2## and 0020 L TRFR3##.

Use the following rules and description:

0020 L TRFR2###

Rule: TRFRP### = 'TRFR2###'.

Description: Frame - Silver

0020 L TRFR3###

Rule: TRFRP### = 'TRFR3###'.

Description: Frame - Red

Press Save to save your changes.

Click SAP to return to the SAP Fiori Launchpad.

Step 7: Create Routing

Task Create a work plan for the new product. **Time** 5 min

Short Description For production, the configurable Deluxe Touring Bike requires a work plan. This is created on the basis of the existing routings of other Deluxe Touring Bikes.

Name (Position) Jun Lee (Production Manager)

To create a routing, use the *Create routing* app in the *Production Planning and Execution* area on the *Variant Configuration* page in the *Production Manager* role.

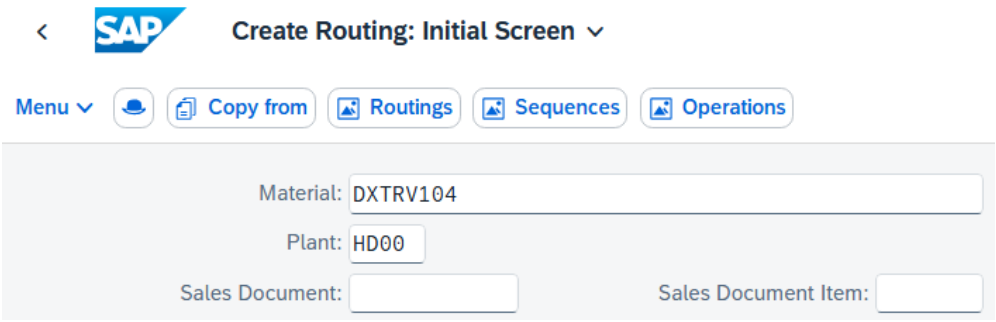
Start



Click on Create.

Enter **DXTRV###** for *material* and **HD00** for *plant*.

DXTRV###
HD00



Press Copy from and select **Routing**. Confirm your selection by clicking on ✓.

Routing

In the Select object to be copied, enter **DXTR1###** for *Material*. The **HD00** *plant* should already be prefilled.

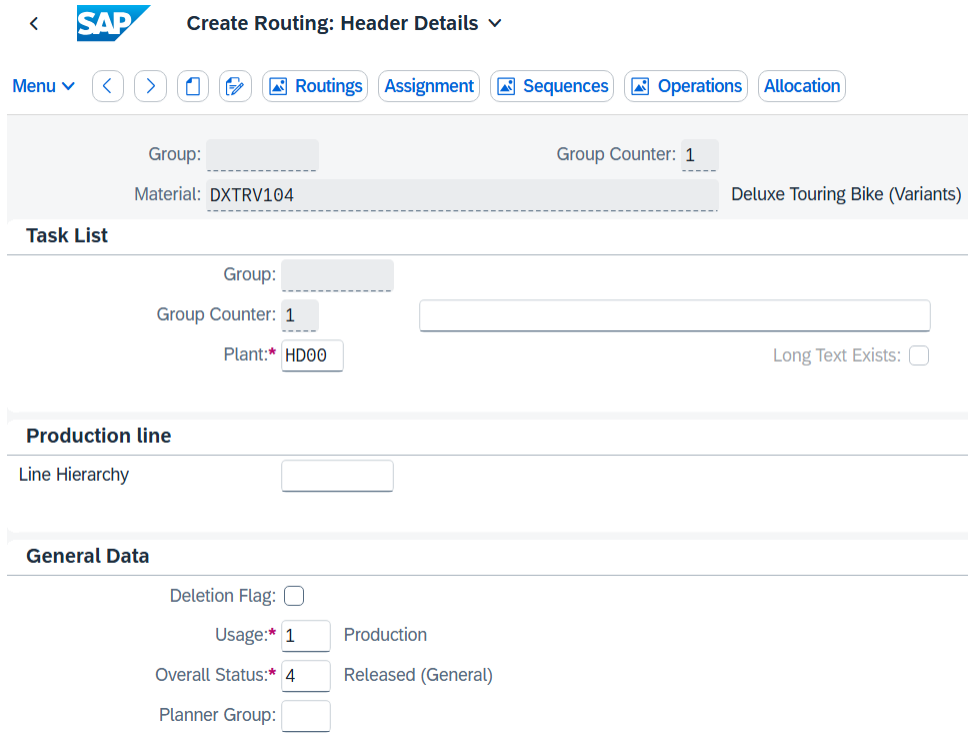
DXTR1###
HD00



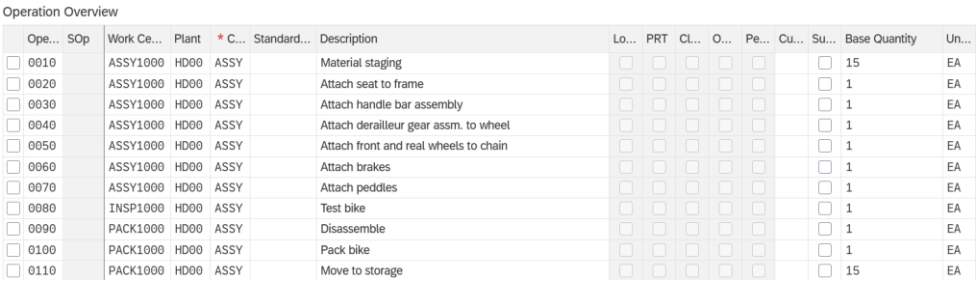
Click on ✓ to continue.

In the header data check, add **4** for *Overall status*, all other entries can be accepted.

4



Press Enter, you will get to the *operation overview*.



Ope...	SOp	Work Ce...	Plant	* C...	Standard...	Description	Lo...	PRT	Cl...	O...	Pe...	Cu...	Su...	Base Quantity	Un...
0010		ASSY1000	HD00	ASSY		Material staging								15	EA
0020		ASSY1000	HD00	ASSY		Attach seat to frame								1	EA
0030		ASSY1000	HD00	ASSY		Attach handle bar assembly								1	EA
0040		ASSY1000	HD00	ASSY		Attach derailleur gear assm. to wheel								1	EA
0050		ASSY1000	HD00	ASSY		Attach front and real wheels to chain								1	EA
0060		ASSY1000	HD00	ASSY		Attach brakes								1	EA
0070		ASSY1000	HD00	ASSY		Attach peddles								1	EA
0080		INSP1000	HD00	ASSY		Test bike								1	EA
0090		PACK1000	HD00	ASSY		Disassemble								1	EA
0100		PACK1000	HD00	ASSY		Pack bike								1	EA
0110		PACK1000	HD00	ASSY		Move to storage								15	EA

Since no changes are necessary here, the work plan can be accepted with a click on Save.

Click SAP to return to the SAP Fiori Launchpad.

Step 8: Manage Production Versions

Task Create a manufacturing version for your configurable Deluxe Touring Bike.

Time 5 min

Short Description After the maximum BOM and routing are created, a production version is created for the configurable Deluxe Touring Bike. This combines the maximum BOM and the routing so that they are used for production.

Name (Position) Jun Lee (Production Manager)

To create a production version, use the *Manage Production Versions* app in the *Production Planning and Execution* area on the *Variant Configuration* page in the *Production Manager* role.

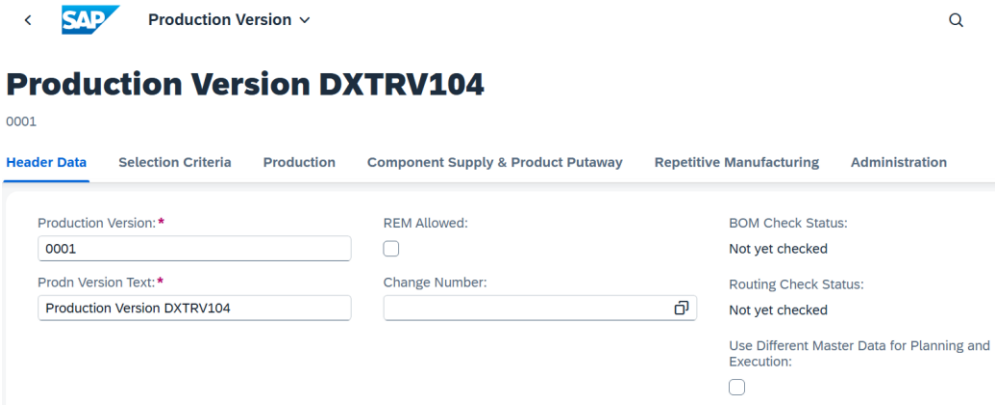
Start



Press Create to create a new production version.

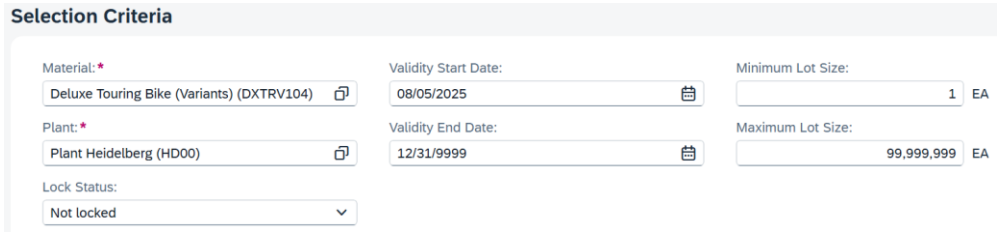
Enter **0001** as *Production version* and **Production Version DXTRV###** for *Prodn Version Text*.

0001
Production Version DXTRV###



In the *Selection Criteria* area, add **DXTRV###** for *Material*, **HD00** for *Plant*, and **1** for *Minimum Lot Size*.

DXTRV###
HD00
1



In the subsequent *Production* area, select **1** as the *Alternative BOM*, **Production** is automatically added for *BOM Usage*.

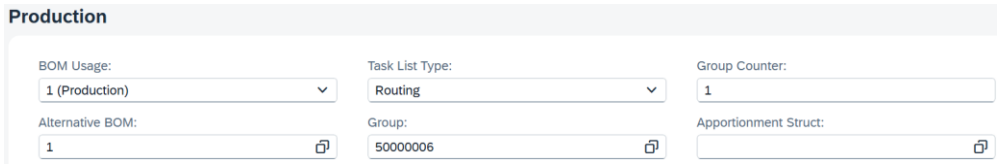
1
Production

Select **Routing** as the *task list type*. For the *task list group*, use the possible entries help to select **your task list group**, the task list *group counter* is automatically supplemented with **1**.

Routing

your task list group

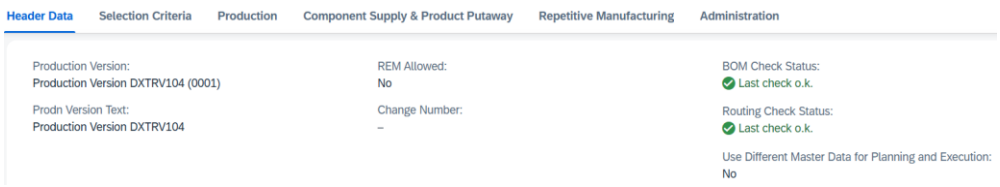
1



Note Loading animations that occur in some fields do not affect the result and you can continue working.

Press Create to save the production version.

Then look again at the header data of your production version. Both the *BOM check status* and the *Routing check status* should now both show *last check OK*.



Click SAP to return to the SAP Fiori Launchpad.

Step 9: Maintain Variant Conditions

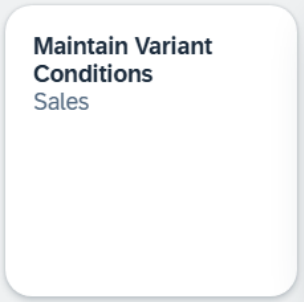
Task Create variant keys. **Time** 5 min

Short Description Variant keys are required so that variant prices can be created. The variant keys are then also needed to link the variant prices with the characteristic values.

Name (Position) Jun Lee (Production Manager)

To create variant keys, use the *Maintain Variant Conditions* app in the *Production Planning and Execution* area on the *Variant Configuration* page in the *Production Manager* role.

Start



Press New Entries to create your own variant closures.

Now enter the following variant key and associated discription:

Variant key	Description
TRFR1###	Touring Bike Frame - Black
TRFR2###	Touring Bike Frame - Silver
TRFR3###	Touring Bike Frame - Red

Note Although the variant keys are named like the associated raw materials, they are independent of each other. However, this makes assignment easier both technically and mentally.

Press Save to save the entries.

Click SAP to return to the SAP Fiori Launchpad.

Step 10: Manage Prices

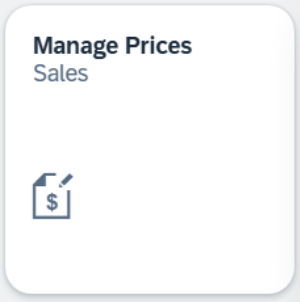
Task Set prices for the product and its variants. **Time** 10 min

Short Description A price is assigned for the configurable Deluxe Touring Bike, but additional variant prices are defined. This calculates the total price from the sum of the product price and the price of the selected feature.

Name (Position) James Shamblin (Head of Sales)

To set sales prices, use the *Manage Prices* app in the *Production Planning and Execution* area on the *Variant Configuration* page in the Head of Sales role.

Start



Press Create to create new condition records.

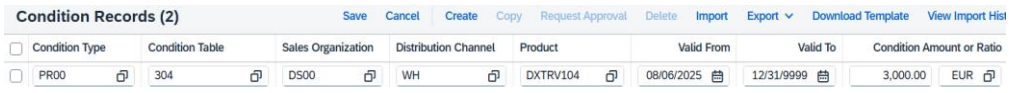
The first step is to create the condition record for the new Deluxe Touring Bike (variants).

Enter **PR00** for *condition type*, **304** for *condition table*, **DS00** for *sales organization*, **WH** for *distribution channel*, and **DXTRV###** as *product*.

PR00
304
DS00
WH
DXTRV###

The *condition amount or ratio* is set at **3,000.00 EUR**.

3,000.00 EUR



Press Save to save the condition record.

Repeat the process to create the condition records for the individual frame colors (the variants). For variant prices, the previously created variant keys are also required.

Use the following table for the individual entries:

Cond. Type[1]	Con.tbl[2]	Variant Key	Sales org[3]	Dis.Channel[4]	Produkt	Amount [5]
VA00	057	TRFR1 ###	DS00	WH	DXTRV ###	100 EUR
VA00	057	TRFR2 ###	DS00	WH	DXTRV ###	60 EUR
VA00	057	TRFR3 ###	DS00	WH	DXTRV ###	80 EUR

[1] = Condition Type

[2] = Condition Table

[3] = Sales Organization

[4] = Distribution Channel

[5] = Condition Amount or Ratio

Press Save to save the condition record.

Click SAP to return to the SAP Fiori Launchpad.

Step 11: Model Variant Configuration

Task Define variant pricing. **Time** 5 min

Short Description After variant prices have been created for the configurable Deluxe Touring Bike, they are linked to the characteristic values via the variant keys.

Name (Position) Jun Lee (Production Manager)

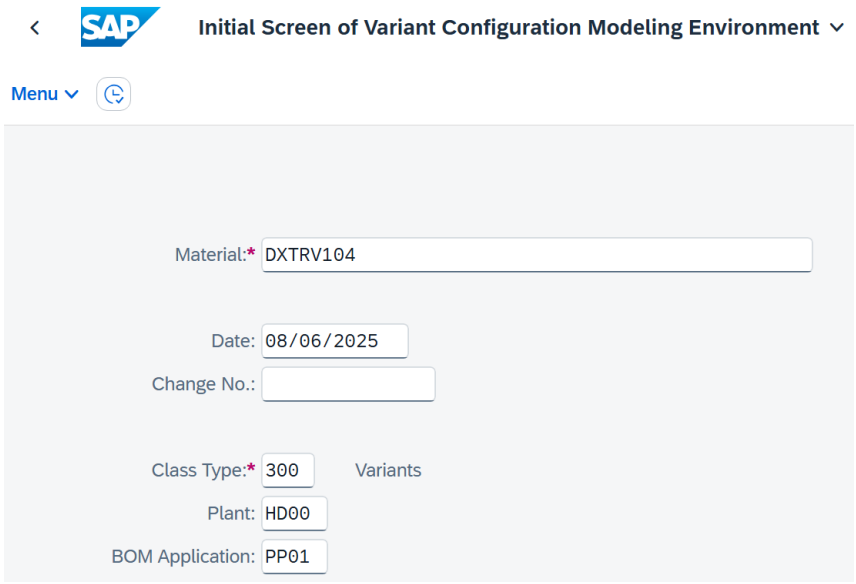
To model variant configuration, use the *VC Modeling Environment* app in the *Production Planning and Execution* area on the *Variant Configuration* page in the *Production Manager* role.

Start



Enter your configurable material **DXTRV###** for *Material*. Check that **300** is selected as the *class type* and enter **HD00** for *Plant*, and **PP01** for *BOM Application*.

DXTRV###
300
HD00
PP01

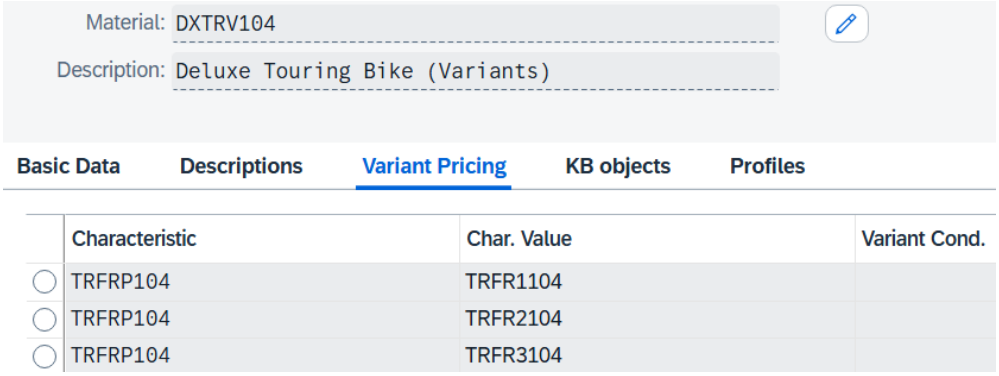


Press to enter the model.

You will be shown the configuration for your configurable material.

Double-click on *DXTRV###* to open the details on the right-hand side of the screen and switch to the *Variant pricing* tab.

To display values in the table, press Expand List in the lower area. You will then see your characteristic TFFRP### and the associated characteristic values.



Press to switch to edit mode. Then enter the variant conditions matching the characteristic values.

Characteristic	Char. Value	Variant Cond.
TRFRP###	TRFR1###	TRFR1###
TRFRP###	TRFR2###	TRFR2###
TRFRP###	TRFR3###	TRFR3###

Press Save to save your changes.

Click SAP to return to the SAP Fiori Launchpad.

Step 12: Create Sales Order

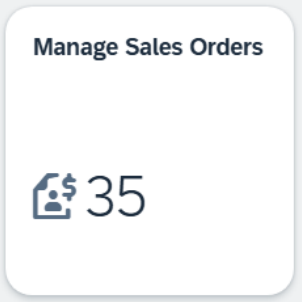
Task Create a sales order. **Time** 10 min

Short Description After the new configurable Deluxe Touring Bike has been fully created, you create a sales order to order the product. Here you will see both the configuration options and the associated prices.

Name (Position) James Shamblin (Head of Sales)

To create a sales order, use the *Manage Sales Orders* app in the *Production Planning and Execution* area on the *Variant Configuration* page in the *Head of Sales* role.

Start



Click Create and select *Create sales order VA01*.

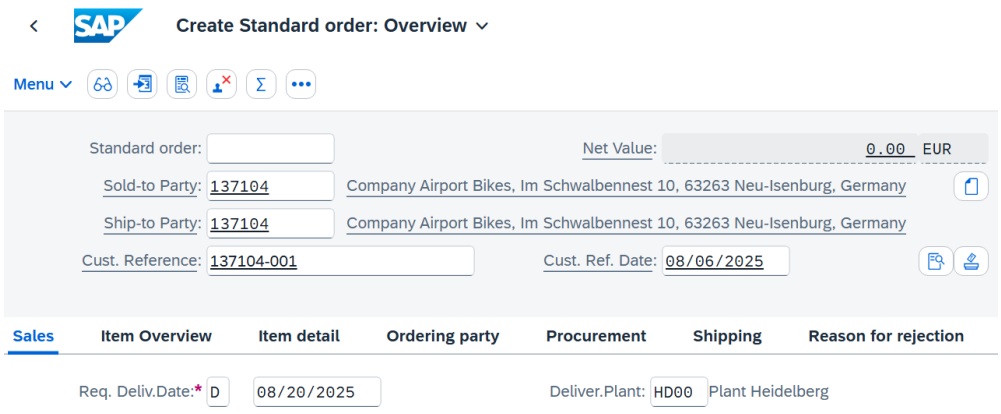
Enter **OR1** to create an Standard Order order and press Continue.

OR1

Enter **137###** for *Sold-to Party* and *Ship-to Party*, **137##-001** as the *Customer Reference*, and the **current date** for *Customer Ref.Date*. In the *Sales* tab, change the *requested delivery date* to **two weeks from today** and enter **HD00** as the *delivery plant*.

137###
137###-001

Current date
Two weeks from today
HD00



Press Enter, a window will open to specify the sales area. Select the combination *DS00 | WH | BI - Germany South / Wholesale / Bicycles*

and click ✓ . The system will automatically add various fields.

On the *Sales* tab, in the *All Items* area, enter **DXTRV###** for *Material* and **10** as the *Order Quantity* in the first line.

DXTRV###
10

All Items

	Item	Material	Req. Segment	Order Quantity
☐		DXTRV104		10

Press Enter. The *Create Standard order: Characteristic Value Assignment* screen appears.

Click on the value help icon in the *Characteristic value* field. Here you can see all available characteristic values and the associated condition.

Frame color

Frame color

S...	Char. Value	Description	Condition
◉		No Entry	
○	TRFR1104	Black	TRFR1104
○	TRFR2104	Silver	TRFR2104
○	TRFR3104	Red	TRFR3104

Select **3 - Red** and press to confirm the selection.

3

Press to perform pricing. This time the condition for the Touring Bike Frame - Red is used for the calculation.

Create Standard order: Characteristic Value Assignment

Menu

Sold-to Party: 137104 Airport Bikes
Material: DXTRV104 Deluxe Touring Bike (schwarz)
Quantity: 10 EA Item: 10
Req. Deliv.Date: 08/20/2025

Characteristic Value Assignment

	Char. Description	Char. Value	Inf...
☐	Frame color	Red	

In this case, the red highlighting does not indicate an error. The process can be continued.

Net Value

Val. 30,800.00 EUR Conditions...

Click Conditions... on to display the condition.



Click on ✓ to close the window and then click .

Press Save to save your order. Confirm the information that the delivery groups consists of one item.

Click SAP to return to the SAP Fiori Launchpad.

Step 13: Create Production Order

Task Create a production order based on the sales order. **Time** 5 min

Short Description Since the configurable Deluxe Touring Bike is a custom build, no quantities are produced in advance. Accordingly, a production order must be created for the order placed.

Name (Position) Lars Iseler (Shop Floor Worker)

To create a production order, use the *Production Order with Sales Order* app in the *Production Planning and Execution* area on the *Variant Configuration* page in the *Shop Floor Worker* role.

Start



Click in the *Sales order* field and select the value help. Enter **137###** for Sold-to Party and press Go. Your sales order is listed. Select the entry and press OK to accept the sales order.

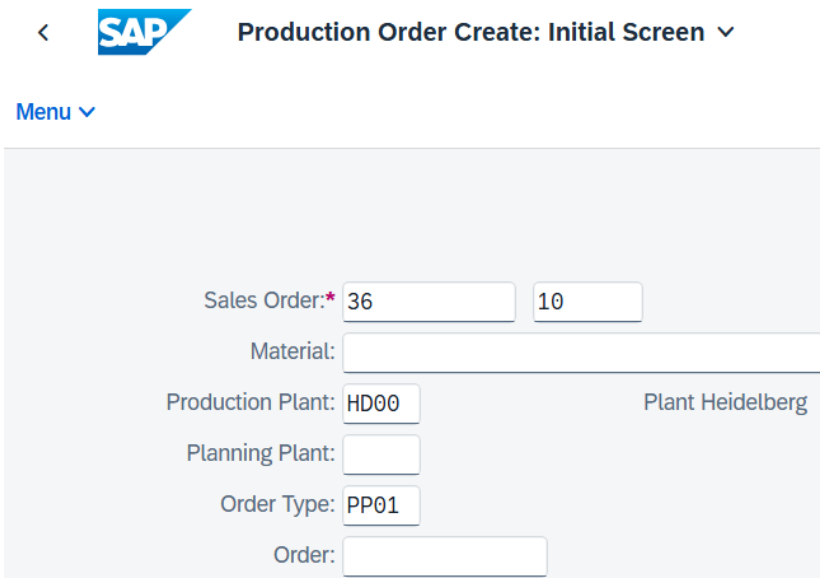
137###

Enter **10** in the second field of the sales order, this is the position in the order.

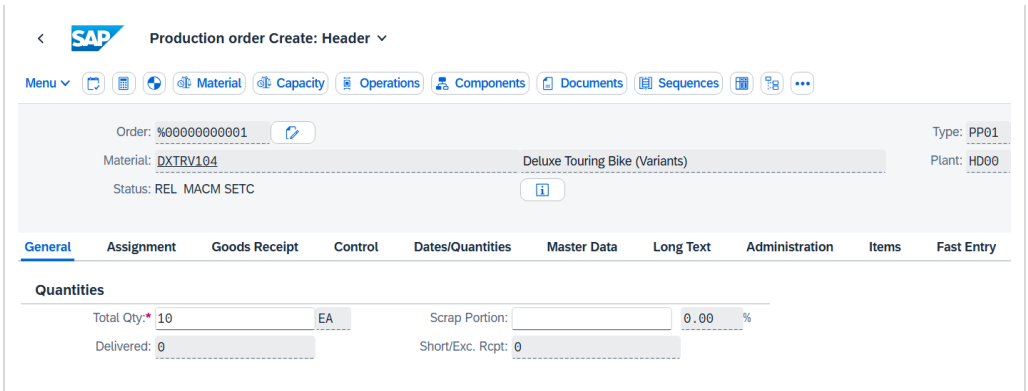
10

Enter **HD00** for *production plant* and **PP01** for *order type*.

HD00
PP01



Press Continue, the Production Order Create: Header screen appears.



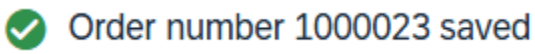
Press Components to display the parts list.

Component Overview

Item	* Component	Description	Reqmt Qty	UoM	It...	Ope...	Seq...	Plant
0010	TRWA1104	Touring Aluminum Wheel Assembly	20	EA	L	0010	0	HD00
0020	TRFR3104	Touring Frame-Red	10	EA	L	0010	0	HD00
0030	DGAM1104	Derailleur Gear Assembly	10	EA	L	0010	0	HD00
0040	TRSK1104	Touring Seat Kit	10	EA	L	0010	0	HD00
0050	TRHB1104	Touring Handle Bar	10	EA	L	0010	0	HD00
0060	PEDL1104	Pedal Assembly	10	EA	L	0010	0	HD00
0070	CHAN1104	Chain	10	EA	L	0010	0	HD00
0080	BRKT1104	Brake Kit	10	EA	L	0010	0	HD00
0090	WDOC1104	Warranty Document	10	EA	L	0010	0	HD00
0100	PCKG1104	Packaging	10	EA	L	0010	0	HD00

As you can see, the maximum BOM has been adjusted to the selected configuration and now contains only one item *0020*, namely the selected *Touring Bike Frame - Red*.

Press Save to save the production order.



Click SAP to return to the SAP Fiori Launchpad.

Step 14: Goods Receipt Materials

Task Post the goods receipt for your production. **Zeit** 10 Min.

Short Description The necessary materials for production must first be procured and posted. Since this is make-to-order production, it should be noted that a special stock must be created. The special stock reserves the materials for the sales order.

Name (Position) Susanne Castro (Goods Receipt Clerk)

To perform the goods receipt, use the *Post Goods Receipt Without Reference* app in the *Production Planning and Execution* area on the *Variant Configuration* page in the *Goods Receipt Clerk* role.

Start



When you start the app, one empty item is already available. Press Create until you have 10 empty items available.

Then add the following values:

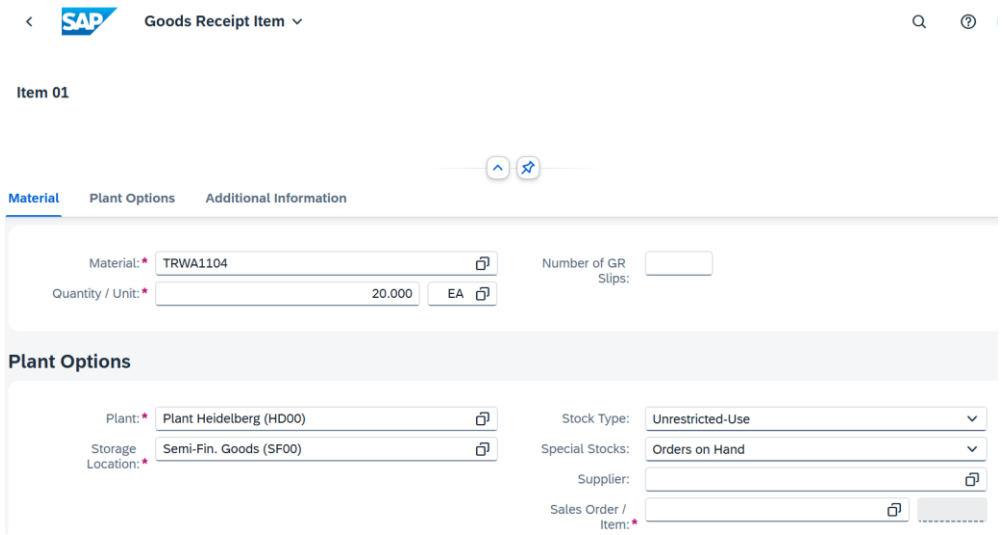
Material	Quantity	Unit	Plant	Storage
TRWA1###	20	EA	HD00	SF00
TRFR3###	10	EA	HD00	RM00
DGAM1###	10	EA	HD00	RM00
TRSK1###	10	EA	HD00	RM00
TRHB1###	10	EA	HD00	RM00
PEDL1###	10	EA	HD00	RM00
CHAN1###	10	EA	HD00	RM00
BRKT1###	10	EA	HD00	RM00
WDOC1###	10	EA	HD00	RM00
PCKG1###	10	EA	HD00	RM00

Note Since the selected strategy provides for make-to-order production, the stock must be shown as order stock. In make-to-order production, only material that is explicitly scheduled for the corresponding order is used.

Click on › at the end of *item 01* to open the details.

In the *Plant Options* section, select **Orders on Hand** for *Special Stocks*.

Orders on Hand



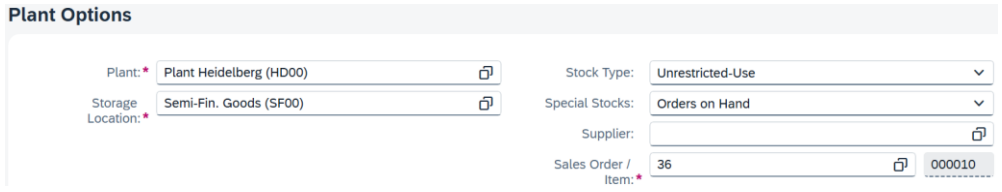
Then use the value help in the *Sales order / Item* field. A window will open.

Geben Sie Ihre Kundennummer **137###** als Suchbegriff ein und drücken Sie Go. Der entsprechende Kundenauftrag wird angezeigt.

137###



Select it, it will be applied automatically.



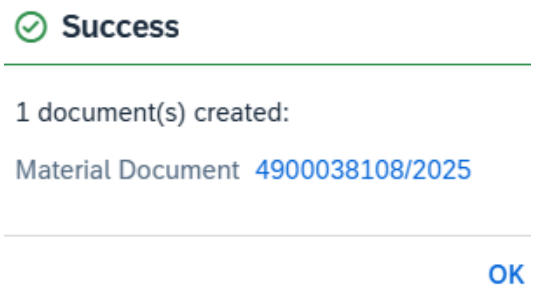
Press Apply, you will return to the position overview.

Repeat the process to show all other items as special stock with reference to the sales order.

Using the Special Stocks column, you can finally check whether all items are shown as Orders on Hand.

	Item	Material	Quantity / Unit		Plant	Storage Location	Stock Type	Special Stocks
☐	01	TRWA1104	20.000	EA	Plant Heidelberg (HD00)	Semi-Fin. Goods (SF00)	Unrestricted-Use	Orders on Hand
☐	02	TRFR3104	10.000	EA	Plant Heidelberg (HD00)	Raw Materials (RM00)	Unrestricted-Use	Orders on Hand
☐	03	DGAM1104	10.000	EA	Plant Heidelberg (HD00)	Raw Materials (RM00)	Unrestricted-Use	Orders on Hand
☐	04	TRSK1104	10.000	EA	Plant Heidelberg (HD00)	Raw Materials (RM00)	Unrestricted-Use	Orders on Hand
☐	05	TRHB1104	10.000	EA	Plant Heidelberg (HD00)	Raw Materials (RM00)	Unrestricted-Use	Orders on Hand
☐	06	PEDL1104	10.000	EA	Plant Heidelberg (HD00)	Raw Materials (RM00)	Unrestricted-Use	Orders on Hand
☐	07	CHAN1104	10.000	EA	Plant Heidelberg (HD00)	Raw Materials (RM00)	Unrestricted-Use	Orders on Hand
☐	08	BRKT1104	10.000	EA	Plant Heidelberg (HD00)	Raw Materials (RM00)	Unrestricted-Use	Orders on Hand
☐	09	WDOC1104	10.000	EA	Plant Heidelberg (HD00)	Raw Materials (RM00)	Unrestricted-Use	Orders on Hand
☐	10	PCKG1104	10.000	EA	Plant Heidelberg (HD00)	Raw Materials (RM00)	Unrestricted-Use	Orders on Hand

Press Post to perform the goods receipt.



Click SAP to return to the SAP Fiori Launchpad.

Step 15: Post Goods Movement

Task Post the goods movement for the production order. **Time** 5 min

Short Description After the materials for the production order are available in the warehouse, they can be booked out for the production order.

Name (Position) Sanjay Datar (Warehouse Employee)

To transfer the raw materials to the production order, use the *Post Goods Movement* app in the *Production Planning and Execution* area on the *Variant Configuration* page in the *Warehouse Employee* role.

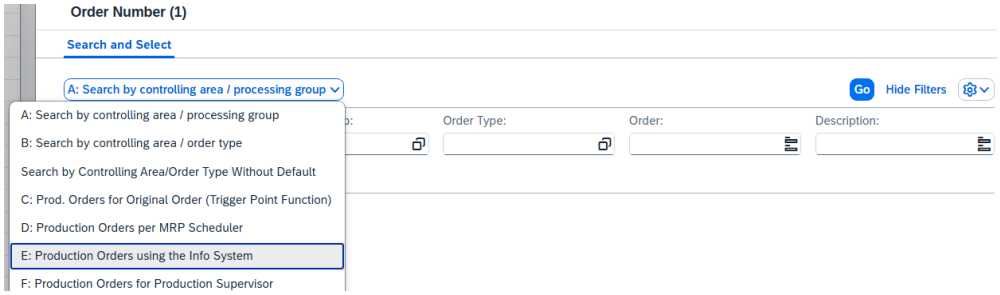
Start



Select **A07 Goods issue** as *movement type* and **R08 Order**.

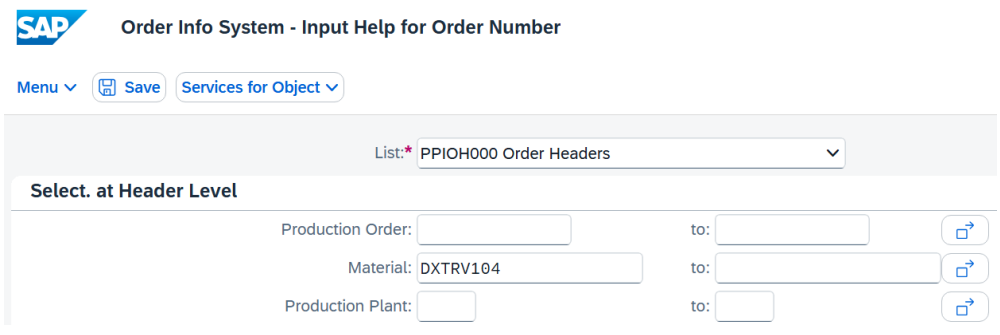
Then click in the third field and click on the value help icon .

Select the entry *E: Production orders using the info system* in the upper dropdown.

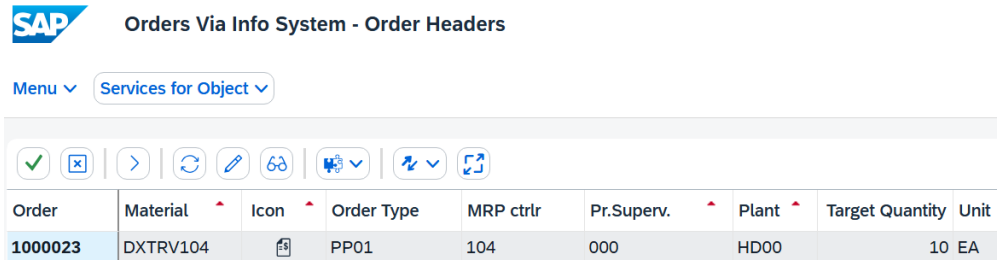


The *Order Info System – Input Help for Order Number* screen opens. Enter **DXTRV###** for *material*.

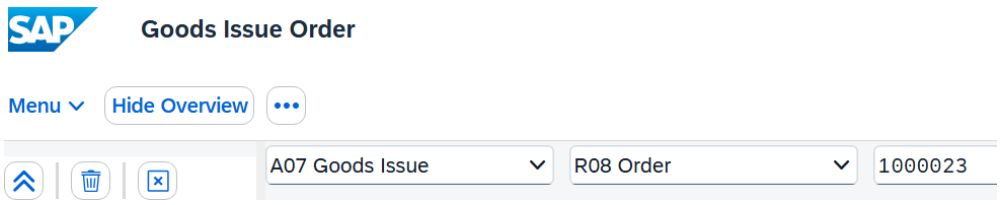
DXTRV###



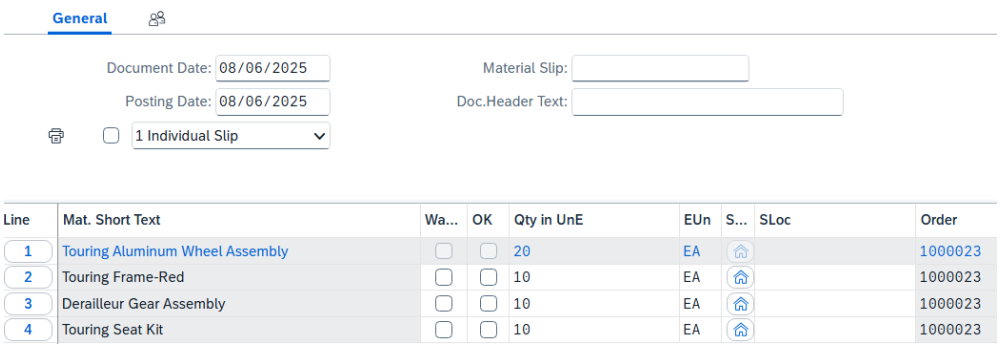
Press Execute, you will be shown your production order.



Double press on the order to accept it. You will automatically return to the *Goods Issue Order* screen.



Press to load the information about the production order.



Press in the lower area to hide the detailed data. You will then get a better overview of all materials and the first line is no longer grayed out.

Line	Mat. Short Text	Wa...	OK	Qty in UnE	EUn	S...	SLoc	Order
1	Touring Aluminum Wheel Assembly			20	EA			1000023
2	Touring Frame-Red			10	EA			1000023
3	Derailleur Gear Assembly			10	EA			1000023
4	Touring Seat Kit			10	EA			1000023
5	Touring Handle Bar			10	EA			1000023
6	Pedal Assembly			10	EA			1000023
7	Chain			10	EA			1000023
8	Brake Kit			10	EA			1000023
9	Warranty Document			10	EA			1000023
10	Packaging			10	EA			1000023

Check **OK** for each destination.

For the material *Touring Aluminum Wheel Assembly* (TRWA1##) enter **SF00** (Semi-Finished Products) and for all other materials enter **RM00** (Raw Materials) for Storage Location.

Line	Mat. Short Text	Wa...	OK	Qty in UnE	EUn	S...	SLoc	Order
1	Touring Aluminum Wheel Assembly	☐	☑	20	EA		SF00	1000023
2	Touring Frame-Red	☐	☑	10	EA		RM00	1000023
3	Derailleur Gear Assembly	☐	☑	10	EA		RM00	1000023
4	Touring Seat Kit	☐	☑	10	EA		RM00	1000023
5	Touring Handle Bar	☐	☑	10	EA		RM00	1000023
6	Pedal Assembly	☐	☑	10	EA		RM00	1000023
7	Chain	☐	☑	10	EA		RM00	1000023
8	Brake Kit	☐	☑	10	EA		RM00	1000023
9	Warranty Document	☐	☑	10	EA		RM00	1000023
10	Packaging	☐	☑	10	EA		RM00	1000023

Press Check to check the document. The status of each line should be highlighted in green. Otherwise, check your entries or whether the stocks have also been correctly posted as special stocks with reference to the sales order.

Line	Status	Mat. Short Text	Wa...	OK	Qty in UnE	EUn	SLoc	Order
1	■	Touring Aluminum Wheel Assembly	☐	☑	20	EA	Semi-Fin. Goods	1000023
2	■	Touring Frame-Red	☐	☑	10	EA	Raw Materials	1000023
3	■	Derailleur Gear Assembly	☐	☑	10	EA	Raw Materials	1000023
4	■	Touring Seat Kit	☐	☑	10	EA	Raw Materials	1000023
5	■	Touring Handle Bar	☐	☑	10	EA	Raw Materials	1000023
6	■	Pedal Assembly	☐	☑	10	EA	Raw Materials	1000023
7	■	Chain	☐	☑	10	EA	Raw Materials	1000023
8	■	Brake Kit	☐	☑	10	EA	Raw Materials	1000023
9	■	Warranty Document	☐	☑	10	EA	Raw Materials	1000023
10	■	Packaging	☐	☑	10	EA	Raw Materials	1000023

Press Post to perform the goods issue.

Click SAP to return to the SAP Fiori Launchpad.

Step 16: Confirm Production Order

Task Confirm the production order. **Zeit** 5 Min.

Description After the Deluxe Touring Bike is produced in the desired configuration, its completion is reported.

Name (Position) Lars Iseler (Shop Floor Worker)

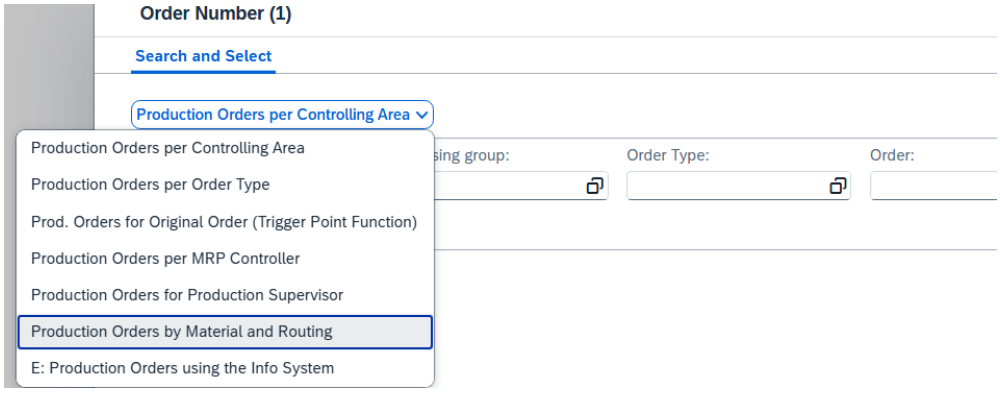
To confirm a production order, use the *Enter Production Order Confirmation* app in the *Shop Floor Worker* role on the *Variant Configuration* page in the *Production Planning and Execution* area.

Start



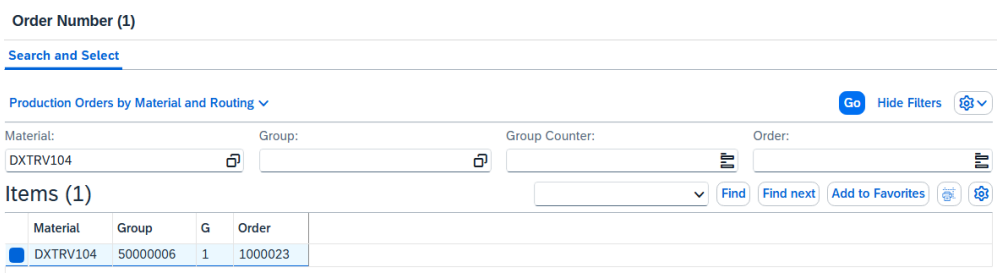
Click in the *Order* field and press the value help icon .

Select the *Production orders by material and routing* entry in the upper dropdown.



Enter **DXTRV###** for *material* and press Go. Your production order is displayed.

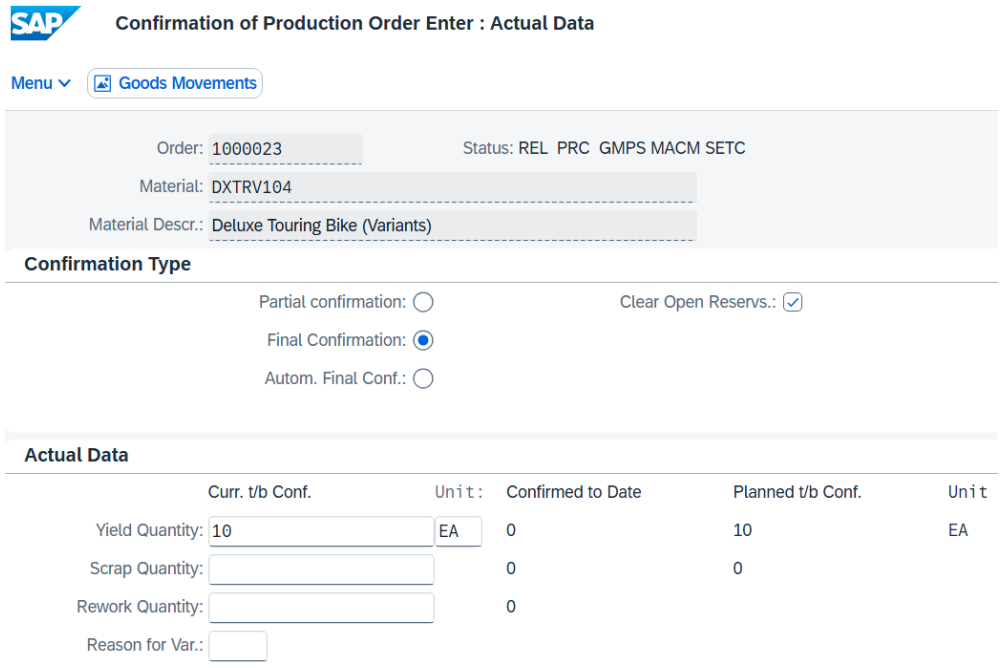
DXTRV###



Select the order and press OK.

Back in the *Enter production order Confirmation : Initial screen*, press Enter.

The displayed settings can be accepted without further adjustments.



Press Save to capture the confirmation.

Click SAP to return to the SAP Fiori Launchpad.

Step 17: Post Goods Receipt for Production Order

Task Post the goods receipt from the production order . **Time** 5 min

Short Description After the production order has been completely confirmed, you post the manufactured Deluxe Touring Bikes to the stock of goods. This stock is also tied to the sales order.

Name (Position) Susanne Castro (Goods Receipt Clerk)

To enter the goods receipt for the production order, use the *Post Goods Receipt for Production Order* app in the *Production Planning and Execution* area on the *Variant Configuration* page in the *Goods Receipt Clerk* role.

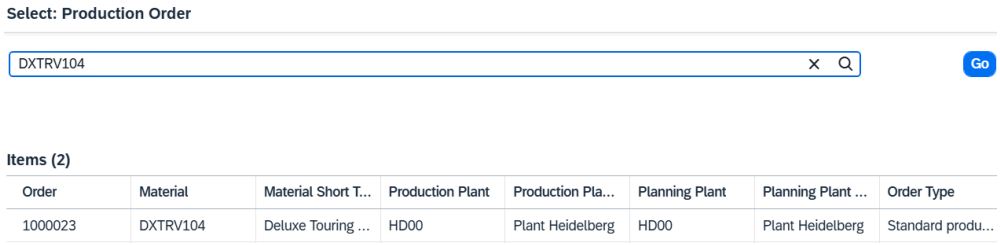
Start



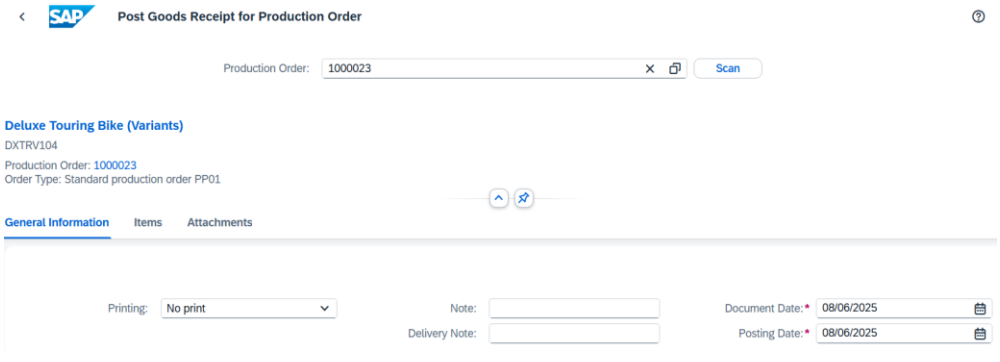
Press the value help icon in the *Production Order* field.

In the window that opens, enter **DXTRV###*** and press Go to determine your order.

DXTRV###*

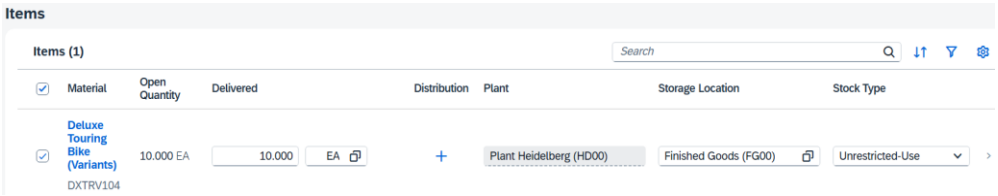


Click on the line to accept the order. The information about the goods receipt is displayed.



Add the missing storage location in the *Items* section. Enter **Finished Goods (FG00)** for *storage location*.

FG00



Press Post to make the goods receipt. You will receive a success message which you can confirm with OK .

Click SAP to return to the SAP Fiori Launchpad.